NON-MF GROUPS EXIST

ANONYMOUS

ABSTRACT. A countable discrete group is MF if it embeds in the unitary group of a norm matrix corona $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$. We prove that non-MF groups exist, resolving in the negative the longstanding question of whether every countable group is MF. Our principal example is an explicit finitely presented sofic group.

The proof follows the central-involution strategy of Bachner–Dogon–Lubotzky. The new ingredient is a one-sided Kazhdan transport theorem: in every operator-norm asymptotic representation, one-sided conjugation preserves the Hilbert–Schmidt asymptotic commutant of a property-(T) subgroup. This forces every homomorphism into the unitary group of a norm matrix corona to map a distinguished central involution to the identity, while an auxiliary Clifford quotient shows that the involution is nontrivial.

We also prove a tensor-power amplification lemma for operator-norm models. It implies that MF is closed in the space of marked groups and under directed colimits; consequently, a non-MF group exists if and only if a finitely generated, or finitely presented, one does. Further consequences include a separable exact stably finite reduced group C^*-algebra that is not MF and a hyperlinear trace that is not MF.

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Contents

1. Introduction

The *MF problem* (MF = matricial field) asks whether every countable discrete group admits finite-dimensional matrix models whose multiplicative errors tend to zero in operator norm and which asymptotically separate every nontrivial element from the identity [Sh]. The separation clause is part of the definition, as constant identity maps are asymptotically multiplicative for every group. Throughout, MF for groups means the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE]; stronger conventions are discussed after Proposition 2.3. To our knowledge no counterexample was previously known [Sc, Sh]. We construct an explicit eight-generator, forty-one-relator group that is not MF. The presentation uses separate generators for the three translations and the three generators of the linear action. The Tietze reduction in Section 7 gives a six-generator, thirty-two-relator presentation.

By tensor-power amplification (Lemma 2.4), the element-dependent separation in this definition can be made uniform. Hence MF is closed in the space of marked groups and under directed colimits (Theorem 2.5), and a non-MF group exists if and only if a finitely generated, or finitely presented, one does (Section 2).

Residually finite and LEF groups are MF [CDE]. Amenable groups are MF because their reduced group C^*-algebras embed into the universal UHF algebra [Sc20, Theorem B], using the quasidiagonality of nuclear C^*-algebras [TWW].

The word "MF" names several related approximation properties. We use the Carrión–Dadarlat–Eckhardt group property: an embedding of the abstract group into a norm matrix corona [CDE]. Schafhauser's group models additionally approximate the left regular representation [Sc], and the strong models of Gao–Kunnawalkam Elayavalli–Manzoor–Patchell recover the regular norm [GKEMP]. Shulman's amalgamated-product results concern, in particular, the MF property of full group C^*-algebras [Sh]. These stronger group and C^*-algebraic properties imply the group property used here.

The result.

Theorem A (an explicit finitely presented non-MF group)**.** *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$\hat{c} = tct^{-1}, \qquad u = [\,\hat{c}, v_1], \qquad w = u^2 = [\,\hat{c},\, v_1\hat{c}v_1^{-1}].$$

The element w is central and satisfies $w^2 = 1$.

(1) $w \neq 1$ in E, so w is a central involution;

(2) for every sequence (d_n) of positive integers and every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.

In particular E is not MF, and neither $C^*_{\max}(E)$ nor $C^*_{\mathrm{r}}(E)$ is an MF C^*-algebra.

The construction also yields a torsion-free finitely presented non-MF group (Theorem S10.1). The MF residual of the explicit group is computed exactly in Section S6.1.

Outline of the proof. The overarching strategy goes back to Bachner–Dogon–Lubotzky [BDL, Proposition 1.5]: obstruct MF by proving that a distinguished central involution cannot be separated from the identity by operator-norm asymptotic representations. Their argument compresses to the involution's (-1)-eigenspace and invokes operator–Hilbert–Schmidt stability. The new ingredient here is a mechanism that supplies the required Hilbert–Schmidt rigidity directly from one-sided conjugation of a property-(T) subgroup, without assuming operator–Hilbert–Schmidt stability of the ambient group.

Multiplicative defects are measured in operator norm; asymptotic-commutant estimates are measured in normalized Hilbert–Schmidt norm. Let Γ have property (T), let $\iota\colon \Gamma \to H$ be a homomorphism, and let $t, c \in H$ satisfy

$$t\,\iota(\Gamma)\,t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Conjugation by the unitaries in an operator-norm asymptotic representation (U_n) of H defines an asymptotic representation on the Hilbert–Schmidt spaces $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$. This is where the operator norm enters: it controls the multiplicative defects of the conjugation actions. Property (T) produces the projection onto the $\iota(\Gamma)$-invariant vectors. The subgroup inclusion implies that this projection is dominated by its conjugate under t. The two projections are equivalent in a finite algebra and therefore equal. Consequently, conjugation by t preserves the asymptotic commutant of $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm (Theorem 3.5).

Put $\hat{c} = tct^{-1}$ and $u = [\,\hat{c}, \iota(a)\,]$ for some $a \in \Gamma$. Since c commutes with $\iota(\Gamma)$, the sequence $(U_n(c))$ lies in that asymptotic commutant. Theorem 3.5 therefore gives $\|[U_n(\hat{c}), U_n(\iota(\gamma))]_-\|_2 \to 0$ for every $\gamma \in \Gamma$. Consequently, in every operator-norm asymptotic representation (U_n) of H,

$$\|U_n(u) - 1\|_2 \longrightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with $\Theta(H)$. Compression to the corresponding matrix corners, equipped with their normalized traces, gives an operator-norm asymptotic representation of H in which the image of w converges to -1 in operator norm. There the display above gives $\|U_n(u)^2 - 1\|_2 \to 0$, while $\|-1-1\|_2 = 2$. Hence $\Theta(w) = 1$, and if $w \neq 1$ then H is not MF (Theorem 4.2). Normalizing the trace on each corner is essential. Enlarging a model by an identity block preserves its operator-norm defects and separations but can make its normalized Hilbert–Schmidt distances arbitrarily small (Section S9).

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α, let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the ascending HNN extension of Γ along α the cosets $t\Gamma$ and $at\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. An auxiliary Clifford quotient witnesses that $w \neq 1$. For a set X, the Clifford group $\mathrm{Cl}(X)$ is generated by a central involution ζ together with involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \neq y$; every permutation of X acts on it by permuting the c_x and fixing ζ (Construction 8.1). Let the ambient group act in this way on $\mathrm{Cl}(X)$ for X the coset space, and send c to the generator c_x at the base coset. Then $\hat{c}$ and $a\hat{c}a^{-1}$ go to the generators at the two distinct cosets, so u goes to a product of two anticommuting involutions and $w = u^2$ to $\zeta \neq 1$ (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I)$$

gives the group E of Theorem A.

Property (T) is used only for the subgroup $\iota(\Gamma)$: the group E itself maps onto $\mathbb{Z}$ by sending t to 1 and every other generator to 0, and therefore does not have property (T). An exact finite-dimensional counterpart of the commutant argument, with equality of dimensions in place of finiteness of the ultraproduct, is Theorem 3.3.

The known implications among the finite approximation properties of countable groups are

$$\begin{array}{ccccc} \text{residually finite} & \Longrightarrow & \text{LEF} & \Longrightarrow & \text{MF} \\ & & \Downarrow & & \\ & & \text{sofic} & \Longrightarrow & \text{hyperlinear.} \end{array}$$

The reverse arrows from LEF to residual finiteness, from MF to LEF, and from soficity to LEF all fail. The finitary alternating group is locally finite, hence LEF, but is infinite simple and therefore not residually finite. Abels' finitely presented group is MF but not residually finite [CDE, Section 2.14 and Corollary 2.18]; since a finitely presented LEF group is residually finite [CDE, Remark 2.11], it is not LEF. Finally, the group E below is sofic but not LEF (Theorems A and D). The reverse of the bottom arrow, hyperlinear $\Rightarrow$ sofic, remains open, as does the additional implication MF $\Rightarrow$ hyperlinear. The canonical trace on $C^*_{\max}(E)$ is hyperlinear but not MF (Theorem E).

Relation to prior work. *MF algebras and traces.* The MF problem for C^*-algebras predates the group question. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^*-algebra was known not to be MF [BK, CDE]. Their natural candidate was $C^*_{\mathrm{r}}(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently implied the existence of abstract stably finite non-MF algebras [JNVWY, FGH]. To our knowledge no countable group was known whose reduced or maximal group C^*-algebra fails to be MF [Sh]. The maximal-algebra obstruction applies to every countable group with a property-(T) subgroup properly conjugated into itself. The corresponding Kazhdan projection is strictly dominated by its conjugate, so the maximal group algebra contains a proper isometry and is not stably finite, and therefore is not MF (Proposition S12.3).

There is a parallel hierarchy for traces. Brown developed tracial analogues of finite-dimensional approximation properties [Br], and MF traces were studied by Rainone–Schafhauser [RS]. We use Shulman's formulation [ShT], in which hyperlinear and MF traces differ by whether the algebraic defects are measured in normalized Hilbert–Schmidt norm or in operator norm. Several classes are known for which every hyperlinear trace is MF; for example, Shulman proves this for cones [ShC]. By contrast, the non-MF traces arising from the failure of the Connes embedding problem do not factor through tracial ultraproducts of matrix algebras and are not hyperlinear [JNVWY, Sc]. Theorem E separates the hyperlinear and MF trace classes.

Metric approximation problems. In his 2018 ICM address, Thom asked for which Schatten p-norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre, Glebsky, Lubotzky, and Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky and Oppenheim [LuO], and the case $p = 1$ by Bachner, Dogon, and Lubotzky [BDL]. Those results left three approximation problems open: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), or MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. Theorem A settles the MF problem negatively; as of August 2026 the hyperlinear problem remains open. The nonsofic and MF arguments both use a property-(T) subgroup Γ satisfying $t\Gamma t^{-1} \subseteq \Gamma$. In the MF setting, conjugation by t preserves the Hilbert–Schmidt asymptotic commutant of Γ whenever the multiplicative defects converge to zero in operator norm.

Related arguments. Recent work on nonsofic groups uses one-sided conjugation of a property-(T) subgroup together with an element centralizing that subgroup [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]. Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. In the present operator-norm setting, the group acts by conjugation on the Hilbert–Schmidt space of each matrix block. Finiteness of the norm ultraproduct implies that the projection onto the $\iota(\Gamma)$-fixed subspace equals its unitary conjugate (Theorem 3.5).

Slofstra's hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. He also considers the elements sent to the identity by every exact finite-dimensional representation and those sent to the identity by every approximate representation [Slo17, Definitions 2.5–2.7].

These subgroups are the exact and approximate analogues of the MF residual defined below.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

An MF embedding of a group C^*-algebra restricts to an MF embedding of the underlying group. The converse is open. An injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$ need not extend to a C^*-algebra embedding $C^*_{\mathrm{r}}(E) \hookrightarrow \mathcal{Q}$. Theorem A rules out both embeddings for E.

2. Matrix quotients

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements $[g,h] = ghg^{-1}h^{-1}$; for operators $[x,y]_- = xy - yx$. Throughout, $*$-homomorphisms are not assumed unital unless explicitly stated; in particular, an MF embedding may be nonunital. For a unital C^*-algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q \in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace. When the trace must be specified, we write $\|x\|_{2,\mathrm{tr}_r}$. The unnormalized trace is $\mathrm{Tr} = r \cdot \mathrm{tr}_r$ on $M_r(\mathbb{C})$. We repeatedly use the inequalities $\|x\|_2 \le \|x\|$; $|\,\mathrm{tr}_r(x)| \le \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, write

$$\mathcal{Q}((d_n)_{n\in\mathbb{N}}) \;=\; \prod_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}), \tag{1}$$

where the product is the bounded C^*-product and $\bigoplus_n$ is the c_0-sum, the ideal of sequences with $\|x_n\| \to 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q\colon \prod_n M_{d_n}(\mathbb{C}) \to \mathcal{Q}$ for the quotient map.

Following Blackadar–Kirchberg [BK], a separable C^*-algebra is *MF* if it embeds into an algebra of the form (1).

Convention on the dimension sequence. When the dimension sequence is not displayed, $\mathcal{Q}$ denotes $\mathcal{Q}((d_n)_{n\in\mathbb{N}})$ for a sequence fixed by the surrounding discussion. If no sequence has been fixed, a map into $\mathcal{Q}$ or $\mathcal{U}(\mathcal{Q})$ is understood to allow the sequence (d_n) to vary; likewise, a quantification over all such maps also ranges over all dimension sequences. Proposition 2.3 shows that nothing changes if the dimensions are required to be strictly increasing.

Three quotients of $\prod_n M_{d_n}(\mathbb{C})$ appear below, differing in which sequences they annihilate:

- $\mathcal{Q}$, where $\|x_n\| \to 0$: the norm matrix corona (1), and the target of an MF embedding;
- B_ω, where $\lim_\omega \|x_n\| = 0$: the norm ultraproduct of Section 3, a finite C^*-algebra, which contains the Kazhdan projection;
- the tracial ultraproduct, where $\lim_\omega \|x_n\|_2 = 0$: the target for hyperlinearity.

The ideals defining $\mathcal{Q}$ and B_ω are given by operator-norm convergence, the third by normalized Hilbert–Schmidt convergence; Section S9 identifies the step that needs the operator norm.

The next two lemmas are standard (cf. [BK, Lo98]); we include the short proofs to fix the constants used later.

Lemma 2.1 (lifting)**.** *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n). Unitarity of $q((x_n))$ gives $\|x_n^* x_n - 1\| \to 0$. Once $\|x_n^* x_n - 1\| \leq \frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n(x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \leq 2\|x_n\| \|x_n^* x_n - 1\| \to 0$ by continuous functional calculus. Set $u_n = 1$ at the finitely many remaining indices. $\square$

Write

$$\mathcal{N} = \{(u_n) \in \textstyle\prod_n \mathcal{U}(d_n) \ : \ \|u_n - 1\| \longrightarrow 0\}, \tag{2}$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient)**.** *For every sequence (d_n), with $\mathcal{N}$ as in (2), the map*

$$\kappa_{(d_n)} \colon \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. Thus a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ may be represented by unitary lifts $U_n(g)$ whose multiplicative defects tend to 0 in operator norm, with two lift sequences identified when their quotient tends to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective. Surjectivity is Lemma 2.1. $\square$

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n \colon H \to \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \to 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \to 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism $\Theta \colon H \to \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

For a finite set $F \subseteq H$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$, an *approximate unitary representation* is a map $V \colon H \to \mathcal{U}(r)$, for some

$r \geq 1$, such that

$$\|V(g)V(h) - V(gh)\| \leq \varepsilon \quad (g, h \in F),$$
$$\|V(g) - V(h)\| \geq \delta \quad (g \neq h \in F).$$

Proposition 2.3 (Equivalent formulations of the MF property)**.** *For every countable group, the group-MF definition above — an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

The nontrivial point in this equivalence is that an element-dependent positive separation in the corona can be made uniform, by the following operator-norm amplification.

Lemma 2.4 (tensor amplification)**.** *Let $u, v \in \mathcal{U}(d)$ and $\|u - v\| \geq \delta > 0$. If $N \in \mathbb{N}$ satisfies $8 < N\delta^2$, then for some $1 \leq p \leq N$,*

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \geq 1.$$

Proof. Put $z = u^*v$. It has a spectral value $e^{2\pi i\alpha}$ with $|1 - e^{2\pi i\alpha}| \geq \delta$. If $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$, then $2\sin(\pi\alpha') \geq \delta$. Successive multiples move by $\pm\alpha'$ modulo 1, so some $p \leq \lceil 1/(6\alpha')\rceil \leq \pi/(3\delta) + 1$ has $\{p\alpha\} \in [1/6, 5/6]$. Since $e^{2\pi ip\alpha}$ belongs to the spectrum of $z^{\otimes p}$,

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \geq |1 - e^{2\pi ip\alpha}| \geq 1.$$

Finally, $\delta \leq 2$ for two unitaries, and $8 < N\delta^2$ therefore gives $\pi/(3\delta) + 1 < N$. ∎

Telescoping the difference of two p-fold tensor products of contractions shows that a pth tensor power multiplies every multiplicative defect by at most p.

Proof of Proposition 2.3. Lemma 2.2 gives the equivalence between a homomorphism to $\mathcal{U}(\mathcal{Q})$ and an operator-norm asymptotic representation. Positive dimensions can be made strictly increasing by replacing the nth model V_n with

$$I_{d_0} \oplus \cdots \oplus I_{d_{n-1}} \oplus V_n.$$

Block-diagonal operator norms are maxima, so this changes neither defects nor separations.

Suppose $\rho\colon G \to \mathcal{U}(\mathcal{Q}((d_n)))$ is injective and choose unitary lifts $u_n(g)$. Fix a finite $F \subseteq G$ and $\varepsilon > 0$. For each $g \neq h$ in F, injectivity supplies $\delta_{g,h} > 0$ and arbitrarily large coordinates with $\|u_n(g) - u_n(h)\| \geq \delta_{g,h}$. Choose $N_{g,h}$ with $8 < N_{g,h}\delta_{g,h}^2$, and choose such a coordinate where every required multiplicative defect is at most $\varepsilon/N_{g,h}$. Lemma 2.4 gives a tensor power $p \leq N_{g,h}$ which separates this pair by at least 1, while every required defect remains at most ε. The block sum of these pair-specific models has defect at most ε and separates every distinct pair in F by at least 1. If $|F| \le 1$, the constant one-dimensional model suffices.

Conversely, choose constant-1 local models along an exhaustion of G with errors tending to zero. Their classes define a homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$; eventual separation by 1 makes it injective. Lemma 2.2 identifies this quotient with $\mathcal{U}(\mathcal{Q}((r_n)))$. $\square$

Theorem 2.5 (the MF locus is closed)**.** *For each $k \ge 1$, the MF groups form a closed subset of the space of k-marked groups. Equivalently, every non-MF k-marked group has a finite-radius obstruction: all marked groups agreeing with it on a sufficiently large word ball are non-MF.*

Proof. By Proposition 2.3, failure of MF is witnessed by a finite set and a positive tolerance for which the constant-1 local model does not exist. Choose words representing this finite set and all products appearing in its multiplicativity conditions. The relevant equalities and inequalities among these words determine a basic clopen neighborhood, and the same local obstruction holds throughout it. A word ball containing all of these words gives the finite-radius formulation. $\square$

Directed-colimit permanence. The class of countable MF groups is closed under directed colimits.

Proof. Let $G = \varinjlim_i G_i$ with every G_i MF. Given finite $F \subseteq G$ and $\varepsilon > 0$, choose a stage containing lifts of all elements and products occurring in the local conditions on F, and then a later stage at which every equality and inequality among those finitely many elements has stabilized to its truth value in G. Composing a constant-1 local model at that stage with the lifts gives one with the prescribed data on F. Proposition 2.3 gives that G is MF. $\square$

The amplification step plays the role that rank amplification plays in the direct-limit theorem for linear sofic groups [AP].

Finite-presentation reduction. The following are equivalent:

(1) a countable non-MF group exists;
(2) a finitely generated non-MF group exists;
(3) a finitely presented non-MF group exists.

Proof. If G is non-MF, a failed local condition is witnessed on a finite set $F \subseteq G$. The subgroup $H = \langle F \rangle$ is non-MF, since an MF model of H on that finite data would contradict the obstruction in G.

Write $H = F_k/N$. By Theorem 2.5, a basic cylinder around H, determined by finitely many words $W \subseteq F_k$, consists entirely of non-MF groups. Let R_+ be the words in W which are trivial in H, and put $P = F_k/\langle\langle R_+ \rangle\rangle$. Then P is finitely presented and maps onto H. Every required equality holds in P, while every word of W nontrivial in H remains nontrivial in P because of the quotient map $P \twoheadrightarrow H$. Thus P lies in the same non-MF cylinder. The remaining implications are immediate. $\square$

Two stronger conventions also occur in the literature. The maps in [GKEMP] must recover the canonical trace and the left-regular norm of every group-ring element. The purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. One-sided conjugation in matrix models

One-sided conjugation has analogous consequences in finite groups, finite-dimensional representations, and norm ultraproducts. For a homomorphism π to a finite group, $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$, since the left side is a conjugate of the right side contained in it and the two are finite of the same cardinality. In finite dimensions, the commutant C of the base satisfies $C \subseteq \pi(t)C\pi(t)^{-1}$; equality follows because the two spaces have the same dimension. In a norm ultraproduct, the fixed-space projection satisfies $P \leq VPV^*$ when $L \leq H$ and $tLt^{-1} \subseteq L$, where P is the projection onto the L-fixed vectors and V represents t; the two projections are Murray–von Neumann equivalent, and finiteness of the ultraproduct gives $P = VPV^*$.

The two lemmas of this section are standard facts about finite C^*-algebras; see, e.g., [Bla].

Lemma 3.1 (comparison in a finite algebra)**.** *Let A be a finite unital C^*-algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \qquad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1-q)$, and $\sigma^*\sigma = q + (1-q) = 1$, so σ is an isometry. Since A is finite, σ is unitary, and $\sigma\sigma^* = p + (1-q) = 1$ gives $p = q$. ☐

Lemma 3.2 (finiteness of a norm quotient of matrix blocks)**.** *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_{\mathrm{c}} = \prod_n B(K_n) \Big/ \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0-sum. Then B_{c} is a finite C^*-algebra: every $\sigma \in B_{\mathrm{c}}$ with $\sigma^*\sigma = 1$ satisfies $\sigma\sigma^* = 1$.

Proof. Lift σ to a bounded family (σ_n). Then $\|\sigma_n^*\sigma_n - 1\| \to 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^*\sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary $w_n = \sigma_n(\sigma_n^*\sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \to 0$. Hence σ is the class of (w_n), and $w_n w_n^* = 1$ at every such n gives $\sigma\sigma^* = 1$. ☐

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n, and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$. The same polar-correction argument shows that B_ω

is finite, with ordinary eventual convergence replaced by convergence along ω. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n \xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\operatorname{ran} P \subseteq \operatorname{ran} Q$ makes QP and P act identically on K_ω, so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, $\Gamma \leq H$ a subgroup, $t, c \in H$, and $a \in \Gamma$. Assume that $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi \colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big(\big[\, tct^{-1},\ a(tct^{-1})a^{-1}\big]\big) \;=\; 1.$$

Proof. Let

$$C = \{x \in \operatorname{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\operatorname{End}_k(V)$, and write $\Phi(x) = \pi(t)\, x\, \pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\operatorname{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1} x\, \pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x, and therefore

$$\pi(\gamma)\, y = \pi(t)^{-1}\, \pi(t\gamma t^{-1})\, x\, \pi(t) = \pi(t)^{-1}\, x\, \pi(t\gamma t^{-1})\, \pi(t) = y\, \pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \;=\; \Phi(C) \;=\; \pi(t)\, C\, \pi(t)^{-1}. \tag{3}$$

Now $\pi(c) \in C$ because c centralizes Γ, so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (3). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$ and

$$\pi\big(\big[\, tct^{-1},\ a(tct^{-1})a^{-1}\big]\big) = \big[\, \pi(tct^{-1}),\ \pi(tct^{-1})\big] = 1. \qquad \square$$

Every finite-dimensional linear representation of the group E of Definition 6.1 below, over any field, maps w to the identity; consequently every homomorphism from E to a finite group does as well (Corollary S3.2).

In an MF model, multiplicativity is only asymptotic, so the coordinate fixed spaces need not be invariant. Because the multiplicative defects tend to 0 in operator norm, the adjoint actions on the Hilbert–Schmidt spaces are asymptotically multiplicative, and the corresponding Kazhdan projection belongs to a finite norm ultraproduct, where $P \leq VPV^*$ and $P \sim VPV^*$ imply $P = VPV^*$. The property (T) hypothesis is essential: the group E_{BS} of Section S4 has cyclic base. Its distinguished word is trivial in every finite-dimensional representation over every field, but some homomorphism $E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ maps that word to a nontrivial element.

The asymptotic case. Recall that a *Kazhdan pair* for a property-(T) group L is a finite set $S \subseteq L$ together with a constant $\kappa > 0$ such that, in every unitary representation π of L with no nonzero invariant vector, every unit vector ξ satisfies $\|\pi(a)\xi - \xi\| \geq \kappa$ for some $a \in S$. The next lemma is the spectral-gap form of the Kazhdan projection, which goes back to Akemann and Walter [AW]; see [DN] for a general treatment.

Lemma 3.4 (Kazhdan spectral gap)**.** *Let L be a group with property* (T)*, let $S \subseteq L$ be a finite symmetric Kazhdan set containing* 1 *with Kazhdan constant $\kappa \in (0,1]$, and let π be a unitary representation of L in a unital C^*-algebra. Then the average $h = \frac{1}{|S|}\sum_{a\in S}\pi(a)$ is self-adjoint and*

$$\mathrm{sp}(h) \subseteq \Big[-1,\ 1 - \frac{\kappa^2}{2|S|}\Big] \cup \{1\}.$$

Proof. Symmetry of S makes h self-adjoint, and $\|h\| \leq 1$ puts its spectrum in $[-1,1]$. Let $\mu \in \mathrm{sp}(h)$ with $\mu \neq 1$. Evaluation at μ is a state on $C^*(1,h)$; extend it to a state φ on the ambient C^*-algebra. Let σ_φ be its GNS representation and set $\widetilde{\pi} = \sigma_\varphi \circ \pi$. Since $\varphi((h-\mu)^2) = 0$,

$$\|(\sigma_\varphi(h) - \mu)\xi_\varphi\|^2 = \varphi((h-\mu)^2) = 0.$$

Thus the cyclic vector $\xi = \xi_\varphi$ is a unit vector with $\sigma_\varphi(h)\xi = \mu\xi$. Split $\xi = \xi_0 + \xi_1$ along the $\widetilde{\pi}(L)$-invariant vectors and their orthocomplement, both invariant under $\sigma_\varphi(h)$. On the first summand $\sigma_\varphi(h) = 1$, so $\mu \neq 1$ forces $\xi_0 = 0$. Now

$$1 - \mu = \frac{1}{|S|}\sum_{a\in S}\big(1 - \mathrm{Re}\langle\widetilde{\pi}(a)\xi, \xi\rangle\big) = \frac{1}{2|S|}\sum_{a\in S}\|\widetilde{\pi}(a)\xi - \xi\|^2,$$

and $\widetilde{\pi}$ has no nonzero invariant vector on the orthocomplement, so $\|\widetilde{\pi}(a)\xi - \xi\| \geq \kappa$ for some $a \in S$ and $1 - \mu \geq \kappa^2/(2|S|)$. □

Theorem 3.5 (one-sided conjugation preserves asymptotic commutants)**.** *Let Γ and H be groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, let $t \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma).$$

Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes $\iota(\Gamma)$ in normalized Hilbert–Schmidt norm,

$$\|[x_n, U_n(\iota(\gamma))]_-\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma),$$

then the conjugated sequence does as well:

$$\|[U_n(t)x_nU_n(t)^*, U_n(\iota(\gamma))]_-\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma).$$

Proof. Suppose the conclusion fails: there are $\gamma_0 \in \Gamma$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(t)x_nU_n(t)^*, U_n(\iota(\gamma_0))]_-\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and let each group element act on it by conjugation, $\operatorname{Ad} U_n(g)\, \xi = U_n(g)\, \xi\, U_n(g)^*$. The operator-norm almost multiplicativity of U_n implies that $g \mapsto \operatorname{Ad} U_n(g)$ is an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\,\cdot\,]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, and B_ω is finite: every isometry in it is unitary, by the ultrafilter version of Lemma 3.2. For projections in B_ω, inclusion of ranges in K_ω is equivalent to the projection order. Asymptotic multiplicativity implies that

$$\pi\colon H \longrightarrow \mathcal{U}(B_\omega), \qquad \pi(g) = [\operatorname{Ad} U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Choose a finite symmetric Kazhdan set $S \subseteq \Gamma$ containing 1, with Kazhdan constant $\kappa \in (0, 1]$, put $h = \frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$, and let $\mathrm{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\iota(\gamma))$. By Lemma 3.4, 1 is isolated in $\mathrm{sp}(h)$, so the spectral projection P of h at 1 belongs to B_ω by continuous functional calculus; since $h = 1$ exactly on Fix, its range is Fix.

The two projections agree. Let $V = \pi(t)$ and $Q = VPV^*$. Since $t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma)$, every vector of Fix is fixed by each $\pi(t\iota(\gamma)t^{-1})$; and η is fixed by all $\pi(t\iota(\gamma)t^{-1})$ exactly when $V^*\eta \in \mathrm{Fix}$. Hence $\operatorname{ran} P = \mathrm{Fix} \subseteq V\,\mathrm{Fix} = \operatorname{ran} Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [x_n]_\omega \in K_\omega$; the uniform operator-norm bound makes this class well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\iota(\gamma))$ fixes ξ. Thus $\xi \in \mathrm{Fix}$. Since $Q = P$, we obtain $V\xi \in V\,\mathrm{Fix} = \operatorname{ran} Q = \operatorname{ran} P = \mathrm{Fix}$. The matrix $U_n(t)x_nU_n(t)^*$ is the vector $\operatorname{Ad} U_n(t)\, x_n$ of K_n, so, again by unitary invariance,

$$\|[U_n(t)x_nU_n(t)^*, U_n(\iota(\gamma))]_-\|_2 = \|(\operatorname{Ad} U_n(\iota(\gamma)) - 1)\ \operatorname{Ad} U_n(t)\, x_n\| \longrightarrow 0$$

along ω, for every $\gamma \in \Gamma$. For $\gamma = \gamma_0$ this contradicts $I \in \omega$. □

Corollary 3.6 (subgroup form)**.** *Let $L \leq H$ have property* (T)*, let $t \in H$ satisfy $tLt^{-1} \subseteq L$, and let (U_n) be an operator-norm asymptotic representation of H. If a uniformly bounded sequence (x_n) asymptotically centralizes L in normalized Hilbert–Schmidt norm, then $(U_n(t)x_nU_n(t)^*)$ does as well.*

Proof. Apply Theorem 3.5 to the inclusion $L \hookrightarrow H$. □

4. The central involution obstruction

This section follows the central-involution strategy of Bachner–Dogon–Lubotzky [BDL, Proposition 1.5]. In their setting, operator–Hilbert–Schmidt stability replaces the compressed asymptotic representation by nearby genuine finite-dimensional representations, which map the involution to the identity. Here Theorem 3.5 instead forces the word whose square is the involution to be Hilbert–Schmidt close to the identity on the compressed corner.

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, and $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection on whose corner $q\mathcal{Q}q$ the compression of $\Theta(w)$ is $-q$ exactly. The argument below, and the generalizations in Section S2, compress a matrix model of H to the corners cut out by coordinate lifts of such a projection.

Lemma 4.1 (compression to an asymptotically central corner)**.** *Let H be a group, let $V_{g,n} \in \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H, and let $q_n \in M_{d_n}(\mathbb{C})$ be nonzero projections with*

$$\|[q_n, V_{g,n}]_-\| \longrightarrow 0 \qquad (g \in H).$$

Write $r_n = \operatorname{rank} q_n \geq 1$ and identify $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. Then there are unitaries $W_{g,n} \in \mathcal{U}(r_n)$ with

$$\|W_{g,n} - q_n V_{g,n} q_n\| \longrightarrow 0 \qquad (g \in H),$$

and $(W_{g,n})$ is an operator-norm asymptotic representation of H on the corners.

Proof. Fix g, abbreviate $e = q_n$, $a = V_{g,n}$ and $\delta = \|[e,a]_-\|$, and note that the unit of the corner is e. From $eae - ae = [e,a]_- e$ and $ea^* - a^*e = -([e,a]_-)^*$ we get $\|eae - ae\| \leq \delta$ and $\|ea^* - a^*e\| \leq \delta$, so

$$\|(eae)^*(eae) - e\| = \|ea^*(ea - ae)e\| \leq \delta,$$

since $a^*a = 1$. Once $2\delta \leq \frac{1}{2}$ the corner element eae is invertible in $M_{r_n}(\mathbb{C})$. Its polar correction $W_{g,n} = eae\left((eae)^*(eae)\right)^{-1/2}$ is unitary with $\|W_{g,n} - eae\| \leq 4\delta$ by the estimate in the proof of Lemma 2.1. Set $W_{g,n} = 1$ at the finitely many remaining indices. For $g, h \in H$ write $b = V_{h,n}$ and $\varepsilon = \|ab - V_{gh,n}\|$. Then

$$\|(eae)(ebe) - eV_{gh,n}e\| \leq \|ea(eb - be)e\| + \|e(ab - V_{gh,n})e\| \leq \|[e,b]_-\| + \varepsilon,$$

which tends to 0. Adding the three polar corrections gives

$$\|W_{g,n}W_{h,n} - W_{gh,n}\| \longrightarrow 0. \qquad \square$$

The rounding, compression and polar-correction estimates of Lemma 4.1 follow Bachner–Dogon–Lubotzky [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. Compression to a negative eigenspace appears in Slofstra–Vidick [SV, proof of Proposition 2.7]. The finite normal case, Theorem S2.3, uses the character-isotypic corner of [BDL, proof of Proposition 1.6] together with one-sided conjugation of a property-(T) subgroup.

Theorem 4.2 (central involution obstruction)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Put $\hat{c} = tct^{-1}$ and $u = [\hat{c}, \iota(a)]$ for some $a \in \Gamma$. If $w := u^2$ is a nontrivial central involution of H, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1)*-spectral corner.* Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, we have $\left\|V_{w,n}^2 - 1\right\| \to 0$ and hence $\left\|V_{w,n}^* - V_{w,n}\right\| \to 0$. Set

$$a_n = \tfrac{1}{2}(V_{w,n} + V_{w,n}^*), \qquad e_n = \tfrac{1}{2}(1 + a_n).$$

Then $e_n = e_n^*$, $\|e_n - p_n\| \to 0$, and $\|e_n^2 - e_n\| \to 0$. The spectrum of e_n therefore concentrates near $\{0, 1\}$, and spectral rounding at $\frac{1}{2}$ gives projections at operator-norm distance $o(1)$ from p_n. Since w is central, these projections asymptotically commute with every lift of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Lemma 4.1 now gives an operator-norm asymptotic representation $(W_{g,n})$ of H on the corners $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. On those corners $q_n V_{w,n} q_n = -q_n + o(1)$, because q_n is the complement of a projection within $o(1)$ of $\frac{1}{2}(1 + V_{w,n})$, so

$$\|W_{w,n} + 1\| \longrightarrow 0.$$

By the standing corner-trace convention, all estimates below use tr_{r_n} and are independent of the ratio r_n/d_n.

The commutant on the corner. Since c centralizes $\iota(\Gamma)$, the sequence $(W_{c,n})$ centralizes $(W_{\iota(\gamma),n})$ asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.5 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{\hat{c},n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \to 0$, and squaring gives $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$.

The contradiction. The relation $w = u^2$ gives $\left\|W_{u,n}^2 - W_{w,n}\right\| \to 0$, while $W_{w,n} \to -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\left\|W_{u,n}^2 + 1\right\|_2 \to 0$ and $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$. The triangle inequality would then give $\|2 \cdot 1\|_2 \to 0$, contradicting $\|2 \cdot 1\|_2 = 2$. Thus $\Theta(w) = 1$. Since $w \neq 1$, every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ has nontrivial kernel, so H is not MF. $\square$

Theorem S2.3 replaces $\{1, w\}$ by any finite normal subgroup of the normal closure of the commutators $[tct^{-1}, \iota(\gamma)]$, $\gamma \in \Gamma$. Replacing the two-element average by a Kazhdan spectral projection gives the analogous result for normal property-(T) subgroups, including infinite, noncentral, and torsion-free ones (Theorem S2.7).

5. The basic non-MF construction

Theorem 5.1 (the basic construction)**.** *Let Γ be a finitely presented group with property* (T)*, let $\alpha\colon \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Define $E(\Gamma, \alpha, a)$ by adjoining elements t, c and imposing*

$$t\gamma t^{-1} = \alpha(\gamma), \qquad c^2 = 1, \qquad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$\hat{c} = tct^{-1}, \qquad u = [\hat{c}, a], \qquad w = u^2 = [\hat{c}, a\hat{c}a^{-1}],$$

and impose centrality of w. More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \qquad [w, t] = [w, c] = 1.$$

The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \to E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism $E(\Gamma, \alpha, a) \to \mathcal{U}(\mathcal{Q})$ maps w to 1, so $E(\Gamma, \alpha, a)$ is not MF.

Proof. Finite presentability follows from the relative presentation. Start with the free product of Γ and the free group on t, c, and quotient by the finitely many displayed relations for a finite generating set of Γ.

The relation $w^2 = 1$ need not be imposed. Put $b = a\hat{c}a^{-1}$. Both $\hat{c}$ and b are involutions, $u = [\hat{c}, a] = \hat{c}b$, and $w = u^2 = (\hat{c}b)^2$. Since $\hat{c}^2 = 1$,

$$\hat{c}\, w\, \hat{c}^{-1} = (b\hat{c})^2 = w^{-1}. \tag{4}$$

Centrality also gives $\hat{c}w\hat{c}^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, consider the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

and use Britton's lemma to identify Γ with its canonical copy in G.

Let $X = G/\Gamma$ be its coset space. The Clifford group $\mathrm{Cl}(X)$ is generated by a central involution ζ and involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \neq y$ (Construction 8.1). Every permutation of X acts on $\mathrm{Cl}(X)$ by permuting the c_x and fixing ζ. Let G act in this way and map c to c_Γ. Then

$$\hat{c} \longmapsto c_{t\Gamma}, \qquad a\hat{c}a^{-1} \longmapsto c_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \longmapsto (c_{t\Gamma} c_{at\Gamma})^2 = \zeta \neq 1.$$

The sign ζ is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Thus $w = u^2$ is a nontrivial central involution, and by Theorem 4.2 every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, every such homomorphism has nontrivial kernel. $\square$

Finite presentability of Γ is used only to make $E(\Gamma, \alpha, a)$ finitely presented; for the rest, Γ may be any property-(T) group with a proper injective endomorphism.

6. The explicit affine example

Specialize Theorem 5.1 to

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I). \tag{5}$$

The affine group Γ is finitely presented, and it has property (T) because it is a lattice in $\mathbb{R}^3 \rtimes \mathrm{SL}_3(\mathbb{R})$ [BHV, Corollary 1.4.16 and Theorem 1.7.1]; see also [Bu]. Moreover, Section 7 identifies it with the presentation used below. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, and it is not surjective because $e_1 \notin 2\mathbb{Z}^3$. The same observation gives $a \notin \alpha(\Gamma)$. Hence Theorem 5.1 applies. Relative to Γ, the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \;\middle|\; \begin{array}{c} t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \qquad [w, g] = 1 \end{array} \right\rangle,$$

where S_Γ is a finite generating set of Γ, g runs over the generators, and $w = [\,tct^{-1},\, a\,(tct^{-1})\,a^{-1}]$. Substituting a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear part yields the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{array}{c} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle, \tag{6}$$

and the *base*

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{array}{c} \text{the eight relations in (6)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\; xv_2x^{-1} = v_1,\; xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\; yv_2y^{-1} = v_2^{-1}v_3,\; yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\; zv_2z^{-1} = v_1v_3^{-1},\; zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle. \tag{7}$$

Write $\iota \colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below.

Definition 6.1. Use the six base generators and the relations displayed in (7). Define
(8)
$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{array}{c} \text{(B)} \quad \text{all relations in (7)}, \\ \text{(T)} \quad tv_it^{-1} = v_i^2 \ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C)} \quad c^2 = 1, \\ [c, g] = 1 \ \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W)} \quad [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$
where w abbreviates the word $\left[\, tct^{-1},\, v_1(tct^{-1})v_1^{-1}\right]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Centrality of w together with $c^2 = 1$ implies $w^2 = 1$ by (4). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

Figure 1 shows how the relation families enter the obstruction and the nontriviality witness.

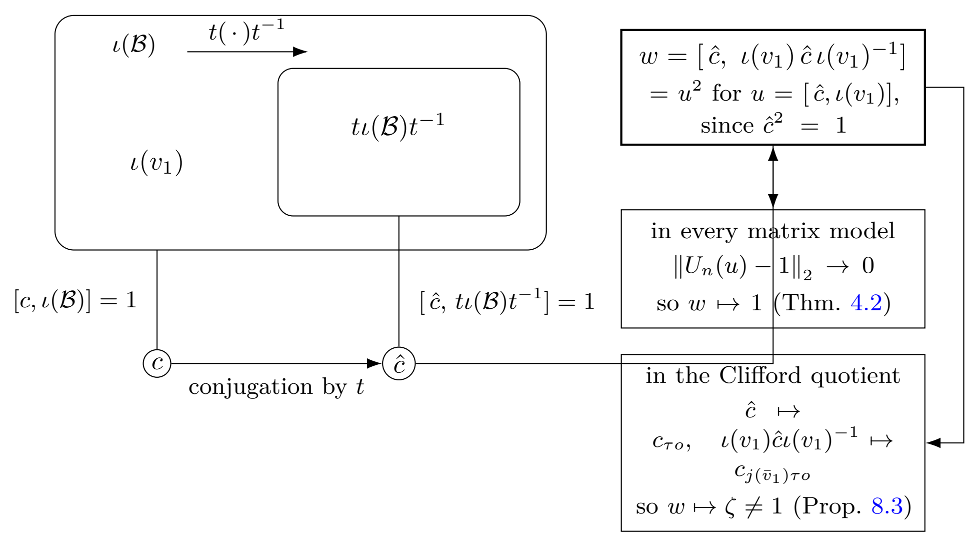

Figure 1: The relations of (8) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[\,\hat{c},\, t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$. Every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ has $\Theta(w) = 1$; the Clifford quotient of Section 8 maps w to $\zeta \neq 1$.

Lemma 6.2 (the distinguished word is a square)**.** *Let H_1 be a group and $x, y \in H_1$ with $x^2 = 1$. Then*

$$[\,x,\ y\,x\,y^{-1}] \ = \ [\,x, y\,]^2.$$

Proof. Put $z = yxy^{-1}$, so $z^2 = yx^2y^{-1} = 1$; thus $z^{-1} = z$ and $x^{-1} = x$. Then $[x, y] = xyx^{-1}y^{-1} = x\,(yxy^{-1}) = xz$, and

$$[x, y]^2 = (xz)^2 = xz\,xz = x\,z\,x^{-1}z^{-1} = [x, z]. \qquad \square$$

Proof of Theorem A. The group E is the specialization of Theorem 5.1 to the data (5), and $w \neq 1$ by Proposition 8.3.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.2) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι, the elements $t, c \in E$, and $a = v_1$. Proposition 7.1 gives property (T) of $\mathcal{B}$, while families (T) and (C) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, respectively. In E we have $\hat{c}^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $x = \hat{c}$ and $y = \iota(v_1)$ gives

$$w = [\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}] = [\,\hat{c}, \iota(v_1)\,]^2 = u^2, \qquad u = [\,\hat{c}, \iota(v_1)\,].$$

The defining relations make w central, and $c^2 = 1$ then implies $w^2 = 1$ by (4). Hence $\Theta(w) = 1$.

Every homomorphism $E \to \mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1, so none is injective and E is not MF. The canonical maps $E \to \mathcal{U}(C^*_{\mathrm{r}}(E))$ and $E \to \mathcal{U}(C^*_{\max}(E))$ are injective, the first because $g \mapsto \lambda_g$ is and the second because the left regular representation factors through $C^*_{\max}(E)$. If $e\colon A \to \mathcal{Q}$ is an injective, possibly nonunital, $*$-homomorphism from a unital algebra, then $u \mapsto e(u) + 1 - e(1)$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(\mathcal{Q})$. Hence an MF embedding of either $C^*_{\mathrm{r}}(E)$ or $C^*_{\max}(E)$ would give an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, a contradiction. $\square$

Corollary 6.3 (uniform finite operator-norm obstruction)**.** *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, for every $d \geq 1$, every map $\phi\colon E \to \mathcal{U}(d)$ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \qquad (g, h \in F_0)$$

satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E, so by Lemma 2.2 its class is a homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, contradicting Theorem A. $\square$

Since E is finitely presented, the forty-one relators of (8) serve as the test set, and the hypothesis becomes a condition on an eight-tuple of unitaries.

Corollary 6.4 (a uniform obstruction on the presentation)**.** *There is a $\delta > 0$ with the following property. For every $d \geq 1$ and every eight-tuple $V_1, \ldots, V_8 \in \mathcal{U}(d)$, if*

$$\|r(V_1, \ldots, V_8) - 1\| \leq \delta \qquad \textit{for each of the forty-one relators } r \textit{ of } (8),$$

then $\|w(V_1, \ldots, V_8) - 1\| < 1$.

Proof. If no such δ exists, then for every n there are a dimension d_n and a tuple $V^{(n)}$ satisfying every relator to within $1/n$ and with $\left\|w(V^{(n)}) - 1\right\| \geq 1$. Each tuple defines a homomorphism $\varphi_n \colon F_8 \to \mathcal{U}(d_n)$, since F_8 is free, and by Lemma 2.2 the classes define a homomorphism $\Phi \colon F_8 \to \mathcal{U}(\mathcal{Q}((d_n)))$. Each of the forty-one relators lies in $\ker \Phi$, its images converging to 1 in operator norm, so Φ factors as $\Theta \circ \pi$ through the quotient map $\pi \colon F_8 \to E$. Theorem A gives $\Theta(w) = 1$, that is $\left\|w(V^{(n)}) - 1\right\| \to 0$, contradicting the lower bound 1. ☐

The compactness proof is ineffective. An explicit value of δ and an explicit test set F_0 would require an effective Kazhdan constant for the base and effective compression estimates.

7. The base and a smaller presentation

The base $\mathcal{B}$ of (7) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) as a lattice in $\mathbb{R}^3 \rtimes \mathrm{SL}_3(\mathbb{R})$ [BHV, Corollary 1.4.16 and Theorem 1.7.1]; see also [Bu].

A six-generator presentation. The group E is generated by the six elements v_1, x, y, z, t, c, and has a Tietze-equivalent presentation on them with thirty-two relators. In (8), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations. The remaining x-action relation follows from $x^3 = 1$. The stable-letter, c-commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c.

After these Tietze transformations, the remaining relations still imply $[\,\hat{c},\, t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$, and the distinguished word $[\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}]$, computed in the same group, is unchanged.

7.1. **Property (T) of the base.**

Proposition 7.1. *The twenty-relator group $\mathcal{B}$ has property* (T)*, in both the real-orthogonal and complex-unitary formulations.*

Proof. By Remark 7.2 the presentation (7) defines $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) [BHV, Corollary 1.4.16 and Theorem 1.7.1]. Let ρ be an orthogonal representation on a real Hilbert space $H_{\mathbb{R}}$. Its complexification $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ has the unitary representation $\rho \otimes 1$. The invariant vectors and displacement norms are those of ρ in each real coordinate, so every Kazhdan pair for the unitary formulation is also one for the orthogonal formulation. ☐

Remark 7.2 (the classical identification). The eight relations (6) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2]; the corresponding matrices are also recorded in [CLV]. Hence $\mathcal{B} \cong \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, and the surjection $\mathcal{B} \to \bar{\Gamma}$ of Lemma 8.2 onto the matrix realization is an isomorphism.

8. An auxiliary Clifford witness

Construction 8.1 (Clifford groups)**.** For a set X, use the real Clifford-algebra convention generated by $(c_x)_{x\in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\mathrm{Cl}(X)$ be the subgroup of units generated by -1 and the c_x. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\mathrm{Cl}(X)$ is a central extension of the elementary abelian 2-group on X, with central kernel generated by ζ. For distinct $x, y \in X$, the commutator $[c_x, c_y]$ equals ζ. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ.

Since $\mathrm{Cl}(X)/\langle\zeta\rangle \cong \bigoplus_X C_2$, every permutation group $\Sigma \curvearrowright X$ gives

$$(\mathrm{Cl}(X) \rtimes \Sigma)/\langle\zeta\rangle \cong \left(\bigoplus_X C_2\right) \rtimes \Sigma.$$

Modulo its sign, $\mathrm{Cl}(X) \rtimes \Sigma$ is thus an ordinary permutational wreath product.

Lemma 8.2 (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

Let $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2, 2, 2, 1)$. The six matrices satisfy all twenty relators of $\mathcal{B}$, and

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z. Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$.

Proof. Each of the twenty relators reduces to an equality between explicit 4×4 rational matrices. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \le \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1 D$ has entry $1/2$ in position $(1,4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. $\square$

Form the direct limit $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ and let τ denote its shift automorphism. Set $\Sigma = T \rtimes \mathbb{Z}$. This is the group G used in Theorem 5.1 for the data (5). Let $j\colon \bar{\Gamma} \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma/j(\bar{\Gamma})$, with base point $o = j(\bar{\Gamma})$. Since $D^{-1}\bar{v}_1 D \notin \bar{\Gamma}$ (Lemma 8.2), we have $\tau o \neq j(\bar{v}_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition 8.3. *$w \neq 1$ in E.*

Proof. Form $W := \mathrm{Cl}(X) \rtimes \Sigma$ as in Construction 8.1. Define a map on the generators of (8) by

$$g \mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

Lemma 8.2 gives the twenty base and six stable-letter relations. The level-zero subgroup fixes o, so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma 8.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W, the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E. $\square$

9. Consequences of the example

Throughout, $\hat{c} = tct^{-1}$ and $u = [\hat{c}, \iota(v_1)]$, so that $w = u^2$. For a countable group G, write $\mathrm{Res}_{\mathrm{MF}}(G)$ for its *MF residual*: the intersection of the kernels of all its homomorphisms to $\mathcal{U}(\mathcal{Q})$ (Definition S5.1). Proposition S5.3 shows that G is MF exactly when this subgroup is trivial. By Theorem A, $1 \neq w \in \mathrm{Res}_{\mathrm{MF}}(E)$.

The quotient by the central involution.

Theorem B (the quotient of E by $\langle w\rangle$)**.** *The quotient $E/\langle w\rangle$ is not MF. The image of $[\iota(v_1), \hat{c}] = u^{-1}$ is nontrivial in $E/\langle w\rangle$ and lies in* $\mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$*; consequently $\{1, w\}$ is a proper subgroup of* $\mathrm{Res}_{\mathrm{MF}}(E)$.

The MF residuals of E and $E/\langle w\rangle$ are computed exactly in Section S6.1. Theorem S6.3 states that if an involution commutes with the conjugate copy of a Kazhdan subgroup and its conjugates under the original subgroup commute pairwise, then every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps all of those conjugates to the same element. In E the pairwise commutators of those conjugates of $\hat{c}$ lie in $\langle w\rangle$, so in $E/\langle w\rangle$ the element $\hat{c}$ satisfies these hypotheses, and every homomorphism $E/\langle w\rangle \to \mathcal{U}(\mathcal{Q})$ maps u to 1. Thus $E/\langle w\rangle$ is a second finitely presented non-MF group.

A stably finite non-MF group algebra. Every MF algebra is stably finite [BK]; the converse fails. The canonical trace on $C^*_{\mathrm{r}}(E)$ is faithful, so $C^*_{\mathrm{r}}(E)$ is stably finite (Lemma S7.1). Abstract examples follow from the failure of the Connes embedding problem [FGH]; the reduced algebra of the finitely presented group E is an explicit one (Theorem C). For $C^*_{\mathrm{r}}(\mathrm{SL}_3(\mathbb{Z}))$ and $C^*_{\mathrm{r}}(\mathrm{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\mathrm{SL}_4(\mathbb{Z})$ [MdlS].

Theorem C (the reduced group algebra of E)**.** *The algebra $C^*_{\mathrm{r}}(E)$ is unital and separable, has a faithful tracial state, and is therefore stably finite, but it is not an MF algebra.*

Proof. $C^*_{\mathrm{r}}(E)$ is unital and separable because E is countable, and by Lemma S7.1 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A. $\square$

The block normal form of Section S7 shows in addition that E, and hence $C^*_{\mathrm{r}}(E)$, is exact. The canonical map $\mathcal{B} \to E$ is injective (Section 8), so E contains an infinite Kazhdan group and is nonamenable. By Lance's theorem, nonamenability of a discrete group is equivalent to nonnuclearity of its reduced group C^*-algebra, so $C^*_{\mathrm{r}}(E)$ is not nuclear [Lan]. In the direction used here, nuclearity is read as the completely positive approximation property: a discrete group whose reduced algebra has that property admits an invariant mean, and E does not. Since $C^*_{\mathrm{r}}(E)$ is nonnuclear, this example does not address the nuclear form of the Blackadar–Kirchberg problem, whether every stably finite separable nuclear C^*-algebra is quasidiagonal.

The maximal algebra behaves differently: $C^*_{\max}(E)$ is not stably finite and has no faithful tracial state. This holds for every group containing a property-(T) subgroup properly conjugated into itself, and comes from a proper isometry built directly from the Kazhdan projection (Proposition S12.3).

A sofic example.

Theorem D (the explicit group is sofic)**.** *The group E of Definition 6.1 is sofic.*

Hence E is hyperlinear. LEF groups are MF [CDE] and E is not MF (Theorem A), so E is not LEF; a residually finite group is LEF, so E is not residually finite either.

Let E_0 be the kernel of the exponent sum in t. For every finite subset of E_0, the T-coordinates lie in a single level of the ascending chain and the lamp coordinates involve finitely many conjugates of c; the subgroup they generate embeds in a residually finite group. Hence E_0 is LEF (Section S7.3). Since E is a split extension of E_0 by $\mathbb{Z}$, it is sofic by the Elek–Szabó amenable-extension theorem [ES06]. This extension does not preserve LEF: LEF groups are MF [CDE], so E_0 is LEF whereas E is not.

MF groups are not closed under extensions or quotients: a Clifford quotient of E is a non-MF extension of an MF group by an MF group, and the MF group F_8 surjects onto E (Section S7, Corollary S5.8).

A hyperlinear non-MF trace. We use Shulman's terminology [ShT]. Hyperlinear and MF traces are defined by the same pointwise bounded, trace-correct matrix models, with the algebraic defects measured in normalized Hilbert–Schmidt norm and operator norm, respectively. The approximating maps themselves need not be linear, positive, unital, or completely positive. Every MF trace is hyperlinear; Shulman asks whether the converse holds.

Lemma 9.1. *Let G be a countable group and let τ_G be the canonical trace on $C^*_{\max}(G)$, determined by $\tau_G(u_g) = 1$ for $g = 1$ and $\tau_G(u_g) = 0$ otherwise. If τ_G is an MF trace, then G is MF.*

Proof. Let $\varphi_n \colon C^*_{\max}(G) \to M_{k_n}(\mathbb{C})$ witness that τ_G is MF and put $\mathcal{Q} = \prod_n M_{k_n}(\mathbb{C})/\bigoplus_n M_{k_n}(\mathbb{C})$, the quotient of the bounded product by the operator-norm c_0 ideal. Pointwise boundedness puts $(\varphi_n(a))_n$ in the bounded product for every a. The operator-norm defect conditions make $\Phi(a) = [(\varphi_n(a))_n]$ a $*$-homomorphism $\Phi\colon C^*_{\max}(G) \to \mathcal{Q}$. Set $p = \Phi(1)$, a projection with $\Phi(C^*_{\max}(G)) \subseteq p\mathcal{Q}p$. Then $\Theta(g) = \Phi(u_g) + (1 - p)$ defines a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q})$. We claim that Θ is injective. If $g \neq 1$ and $\Theta(g) = 1$, then $\Phi(u_g) = \Phi(1)$, so $\|\varphi_n(u_g) - \varphi_n(1)\| \to 0$; the normalized trace is contractive for the operator norm, whereas trace-correctness gives $\mathrm{tr}_{k_n}(\varphi_n(u_g)) \to 0$ and $\mathrm{tr}_{k_n}(\varphi_n(1)) \to 1$, a contradiction. Thus G embeds in the unitary group of a norm matrix corona and is MF (Proposition 2.3). $\square$

Schafhauser [Sc, Definition 1.1 and Proposition 2.2] uses a stronger notion of MF group and relates it to the canonical trace on $C^*_{\mathrm{r}}(G)$. Here we use the Carrión–Dadarlat–Eckhardt notion and the canonical trace on $C^*_{\max}(G)$, as in Lemma 9.1.

Theorem E (a hyperlinear non-MF trace)**.** *The canonical trace τ_E on $C^*_{\max}(E)$ is hyperlinear but not MF. Consequently, there is a separable unital C^*-algebra carrying a hyperlinear non-MF trace. In particular, this gives a negative answer to Shulman's question whether every hyperlinear trace is MF* [ShT].

Proof. By Theorem D, E is sofic. Let $\sigma_n \colon E \to \mathrm{Sym}(d_n)$ be a sofic approximation and $U_n(g)$ the associated permutation unitaries. For a free ultrafilter ω, let

$$\oplus_{2,\omega} M_{d_n}(\mathbb{C}) = \{(x_n) : \lim_{\omega} \|x_n\|_{2,\mathrm{tr}_{d_n}} = 0\}$$

inside the bounded product, and write $\mathrm{tr}_\omega([(x_n)]) = \lim_\omega \mathrm{tr}_{d_n}(x_n)$ for the induced trace on the tracial ultraproduct. The classes $[(U_n(g))_n]$ define a homomorphism

$$E \longrightarrow \mathcal{U}\left(\prod_n M_{d_n}(\mathbb{C})/\oplus_{2,\omega} M_{d_n}(\mathbb{C})\right).$$

Moreover, asymptotic freeness of the sofic approximation gives

$$\mathrm{tr}_\omega([(U_n(g))_n]) = \delta_{g,1},$$

because the normalized trace of a permutation matrix is the proportion of its fixed points. By the universal property of $C^*_{\max}(E)$ this extends to a $*$-homomorphism π with $\tau_E = \mathrm{tr}_\omega \circ \pi$, so τ_E is hyperlinear by Shulman's tracial-ultraproduct characterization [ShT].

If τ_E were an MF trace, Lemma 9.1 would imply that E is MF, contradicting Theorem A. Separability of $C^*_{\max}(E)$ follows from countability of E. $\square$

Supplementary material

The supplementary sections prove general obstruction criteria used above, compute the MF residual of E, and establish further consequences of the construction.

S1. Weighted asymptotic-commutant invariance

Theorem 3.5 holds with an arbitrary weight ν_n in place of the dimension.

Theorem S1.1 (one-sided conjugation at an arbitrary weight)**.** *Let Γ, H, ι, and t be as in Theorem 3.5, let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation, let (ν_n) be nonnegative weights, and let $x_n \in M_{d_n}(\mathbb{C})$ be matrices for which there is C with $\mathrm{Tr}(x_n^* x_n) \le C\,\nu_n$ for all n and, for every $\gamma \in \Gamma$ and every $\varepsilon > 0$, eventually*

$$\mathrm{Tr}\left|x_n - U_n(\iota(\gamma))\,x_n\,U_n(\iota(\gamma))^*\right|^2 \le \varepsilon\,\nu_n.$$

Then $U_n(t)\,x_n U_n(t)^*$ and $U_n(t)^*\,x_n U_n(t)$ satisfy both conditions as well.

Proof. Equip the nth coordinate space with the inner product

$$\langle x, y\rangle_n = \begin{cases} \mathrm{Tr}(y^*x)/\nu_n, & \nu_n > 0,\\ \mathrm{Tr}(y^*x), & \nu_n = 0.\end{cases}$$

If $\nu_n = 0$, the bound $\mathrm{Tr}(x_n^* x_n) \le C\nu_n$ forces $x_n = 0$, so choosing the ordinary Hilbert–Schmidt inner product on that coordinate does not affect the distinguished vector. Conjugation by a unitary is unitary for every one of these inner products. The bound makes (x_n) a bounded ultraproduct vector, and the adjoint actions remain operator-norm almost multiplicative because the scalar normalization of the Hilbert norm does not enter the operator norm. Thus the ultraproduct proof of Theorem 3.5 applies with these weighted coordinate Hilbert spaces. Finiteness of the norm ultraproduct again turns $P \le VPV^*$ into $P = VPV^*$, which gives both conjugation directions. □

Taking $\nu_n = k_n$, the rank of the projection lift in the proof of Theorem S6.1, gives the normalization used there.

S2. Obstructions from a one-sided conjugation datum

By Theorem 3.5, the normal closure N_{conj} of the commutators $[tct^{-1}, \iota(\lambda)]$ is contained in $K_2^\omega(U)$ for every operator-norm asymptotic representation U of H. For a finite normal subgroup of N_{conj}, the corner of Lemma 4.1 is the complement of a projection obtained from its averaging operators (Theorem S2.3), and the central involution of Section 4 is the case $F = \{1, w\}$. For a normal property-(T) subgroup, which may be infinite, noncentral and torsion-free, the corner is a spectral projection of a Kazhdan average (Theorems S2.6 and S2.7). Corollary S2.8 replaces N_{conj} by the subgroup $\mathfrak{D}(H, L)$ of (9), which depends on L alone and not on a choice of t and c.

Definition S2.1 (Kazhdan conjugation datum)**.** Let H be a countable group. A *Kazhdan conjugation datum* in H is a tuple (Λ, ι, t, c), where Λ is a countable group with property (T), $\iota\colon \Lambda \to H$ is a group homomorphism, possibly with nontrivial kernel, and $t, c \in H$, subject to

(M1) $t\,\iota(\lambda)\,t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
(M2) $c\,\iota(\lambda) = \iota(\lambda)\,c$ for every $\lambda \in \Lambda$.

Write $\hat{c} := tct^{-1}$. The datum determines the normal closure

$$N_{\mathrm{conj}} \;=\; \big\langle\!\big\langle\, [\,\hat{c}, \iota(\lambda)\,] \;:\; \lambda \in \Lambda \,\big\rangle\!\big\rangle_H \;\trianglelefteq\; H.$$

For a finitely generated Λ, conditions (M1)–(M2) are imposed by finitely many relations once generating sets are fixed. Conditions (M1)–(M2) imply that c commutes with $\iota(\Lambda)$ and that $\hat{c}$ commutes with the subgroup $t\iota(\Lambda)t^{-1}$.

Definition S2.2 (universal 2-norm triviality)**.** Let $U = (U_n)$ be an operator-norm asymptotic representation of a countable group H and let ω be an ultrafilter refining the cofinite filter. Write

$$K_2^\omega(U) \;=\; \big\{g \in H :\; \lim_\omega \|U_n(g) - 1\|_2 = 0\big\},$$

the kernel of the homomorphism induced by U into the tracial ultraproduct of the matrix blocks along ω, and hence a normal subgroup of H. An element g is *universally* 2*-norm trivial* if $g \in K_2^\omega(U)$ for every such U and every such ω. If $\|U_n(g) - 1\|_2 \to 0$ along the full sequence, then $g \in K_2^\omega(U)$ for every such ω.

Theorem S2.3 (finite-normal obstruction criterion)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Proof. *Universal* 2*-norm triviality of N_{conj}.* Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H (for instance, coordinate unitary lifts of a homomorphism to $\mathcal{U}(\mathcal{Q})$), write $U_{g,n} = U_n(g)$, and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, the sequence $(U_{c,n})$ belongs to the asymptotic commutant of $\iota(\Lambda)$ in normalized Hilbert–Schmidt norm. Theorem 3.5, applied to $x_n = U_{c,n}$, shows that $(U_{\hat{c},n})$ for $\hat{c} = tct^{-1}$ lies in the same asymptotic commutant. Therefore $[\hat{c}, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$; since K_2 is normal, it contains the normal closure of these commutators: $N_{\mathrm{conj}} \subseteq K_2$, and so $F \subseteq K_2$. Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of N_{conj} is universally 2-norm trivial (Definition S2.2).

The coordinate averaging corner. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on F; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent. Since F is normal, (p_n) asymptotically commutes with every lift of $\Theta(H)$. Continuous functional calculus applied to the Hermitian parts yields genuine projections at operator-norm distance $o(1)$ from p_n, and these projections are asymptotically central as well. Let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates and relabel them by $\mathbb{N}$. Lemma 4.1 gives an operator-norm asymptotic representation $(W_{g,n})$ of H on the corners $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$, $r_n = \operatorname{rank} q_n > 0$. Since q_n is the complement of a projection within $o(1)$ of p_n, the finite average vanishes on the corner:

$$(*) \qquad \Big\| \sum_{f \in F} W_{f,n} \Big\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Since $F \subseteq N_{\mathrm{conj}}$ and $(W_{g,n})$ is itself an operator-norm asymptotic representation of H, every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \to 0$. Because F is finite, these convergences are uniform over $f \in F$, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F| 1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ is trivial on F. □

The same conclusion holds for every finite normal subgroup F contained in a subgroup $D \leq H$ whose elements lie in $K_2^{\omega}(U)$ for every operator-norm asymptotic representation U of H and every ω. Finiteness of F is essential: universal 2-norm triviality alone does not force triviality in the corona (Section S9).

The central involution obstruction (Theorem 4.2) is the case $F = \{1, w\}$, where $p_n = \frac{1}{2}(1 + V_{w,n})$; in the corona corner the image of w is -1 exactly, while the finite-stage compressions satisfy $\|W_{w,n} + 1\| \to 0$.

S2.1. **An intrinsic form.** For a subgroup $L \leq H$, let

$$P_L = \{ s \in H : sLs^{-1} \subseteq L \},$$

the *compression semigroup* of L in the terminology of Kun–Thom [KT26], and put $G^+(L) = \langle P_L \rangle$. Then set

$$(9) \qquad \mathfrak{D}(H, L) = \Big\langle\!\Big\langle [gzg^{-1}, \ell] : g \in G^+(L),\ z \in C_H(L),\ \ell \in L \Big\rangle\!\Big\rangle_H ,$$

which depends only on the pair (H, L), not on a choice of datum.

Corollary S2.4 (transport by the compression semigroup)**.** *Let (r_n) and $V_{g,n} \in \mathcal{U}(r_n)$ be an operator-norm asymptotic representation of H, let Λ have property* (T)*, let $\iota\colon \Lambda \to H$ be a homomorphism, and put $L = \iota(\Lambda)$. Suppose that $(x_n \in M_{r_n}(\mathbb{C}))$ is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of L:*

$$\left\|x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^*\right\|_2 \longrightarrow 0 \qquad (\lambda \in \Lambda).$$

Then for every $g \in G^+(L)$, conjugation and inverse conjugation both preserve that asymptotic commutant. Explicitly, both

$$V_{g,n} x_n V_{g,n}^* \quad \textit{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$.

Proof. For a one-sided conjugator s, Theorem 3.5 gives the conclusion for $V_{s,n} x_n V_{s,n}^*$. The equality of the fixed-space projection with its conjugate established in that proof also gives invariance under the inverse conjugation, hence the conclusion for $V_{s,n}^* x_n V_{s,n}$. The set of $g \in H$ for which conjugation by $V_{g,n}$ and by $V_{g,n}^*$ both preserve the asymptotic commutant is a subgroup, and it contains every one-sided conjugator; hence it contains $G^+(L)$. $\square$

Theorem S2.5 (the subgroup $\mathfrak{D}$ and the MF residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \le \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of F to the identity.*

Proof. Let $(U_{h,n})$ be coordinate lifts of an operator-norm asymptotic representation of H and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \to 0\}$, a normal subgroup of H. If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant of L, and by Corollary S2.4 so does $(U_{gzg^{-1},n})$ for every $g \in G^+(L)$. Hence each generator $[gzg^{-1}, \ell]$, $\ell \in L$, displayed in (9) belongs to K_2; since $K_2 \trianglelefteq H$, it contains their normal closure: $\mathfrak{D}(H, L) \le K_2$. Every element of $\mathfrak{D}(H, L)$ is therefore universally 2-norm trivial in every operator-norm asymptotic representation of H. If a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ were nontrivial on a finite normal $F \le \mathfrak{D}(H, L)$, the averaging-corner argument of Theorem S2.3 would contradict this universal 2-norm triviality. $\square$

S2.2. **The normal-Kazhdan form of the obstruction.** A finite group has property (T), and its averaging operator is the projection onto its fixed space. The spectral projection of a Kazhdan average replaces that operator for an arbitrary normal property-(T) subgroup, which may be infinite, noncentral and torsion-free.

Theorem S2.6 (obstruction from universal 2-norm triviality)**.** *Let H be a countable group and let $D \le H$ be a subgroup every element of which is universally* 2*-norm trivial (Definition S2.2). Let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \le D$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Proof. Suppose a homomorphism Θ to $\mathcal{U}(\mathcal{Q})$ is nontrivial on K. Its image $\bar{H} = \Theta(H)$ is countable and, as a subgroup of $\mathcal{U}(\mathcal{Q})$, is MF by definition. Proposition 2.3, applied along an exhaustion of $\bar{H}$, gives an operator-norm asymptotic representation $(V_{g,n})$ in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models provide operator-norm separation, but at growing dimension this yields no lower bound on normalized Hilbert–Schmidt distances. Every operator-norm asymptotic representation of $\bar{H}$ composes with Θ to one of H, so every element of $\Theta(D)$ is again universally 2-norm trivial, while $\bar{K} = \Theta(K) \leq \Theta(D)$ is normal, Kazhdan and nontrivial. We now work in $\bar{H}$.

The Kazhdan corner. Fix a free ultrafilter ω. Let H_ω be the Hilbert-space ultraproduct of the coordinate spaces $\mathbb{C}^{d_n}$, and let $B_\omega = \prod_\omega B(\mathbb{C}^{d_n})$ act on H_ω as in Section 3. The classes $\pi(g) = [V_{g,n}]_\omega$ define a unitary representation of $\bar{H}$ on H_ω. Fix a finite symmetric generating Kazhdan set $S \subseteq \bar{K}$ containing 1, with constant ε_0, put $\theta = 1 - \varepsilon_0^2/(4|S|)$, and let

$$h = \frac{1}{|S|} \sum_{a \in S} \pi(a) \in B_\omega.$$

By Lemma 3.4 the spectrum of h lies in $[-1, 1 - \varepsilon_0^2/(2|S|)] \cup \{1\}$ and hence in $[-1, \theta] \cup \{1\}$, so the spectral projection $P = \chi_{\{1\}}(h)$ belongs to B_ω; its range is $\operatorname{Fix} \pi(\bar{K})$, where $h = 1$. Put $q = 1 - P$.

If $q = 0$ then $\pi(k) = 1$ for every $k \in \bar{K}$, against the operator-norm separation of a fixed nontrivial element of $\bar{K}$. For $g \in \bar{H}$ the projection $\pi(g)P\pi(g)^*$ has range $\operatorname{Fix} \pi(g\bar{K}g^{-1}) = \operatorname{Fix} \pi(\bar{K})$ by normality, so P and q commute with $\pi(\bar{H})$ exactly. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$, nonzero along ω and with $\|[q_n, V_{g,n}]_-\| \to_\omega 0$ for every $g \in \bar{H}$, and lift h to the self-adjoint coordinates

$$h_n = \frac{1}{2|S|} \sum_{a \in S} (V_{a,n} + V_{a,n}^*) \in M_{d_n}(\mathbb{C}),$$

which represent h because S is symmetric. The compression of h to the corner satisfies $qhq \leq \theta q$ in B_ω, and, writing x_+ for the positive part of a self-adjoint x, the positive part is a continuous function of x and the norm ultraproduct computes it coordinatewise, so that inequality is exactly the coordinate statement

$$\left\| (q_n h_n q_n - \theta q_n)_+ \right\| \to_\omega 0. \tag{10}$$

To obtain ordinary convergence before compressing, pass to a subsequence. Enumerate $\bar{H} = \{g_1, g_2, \dots\}$. For each j the set

$$A_j = \left\{ n \ : \ q_n \neq 0, \ \ \max_{i \leq j} \|[q_n, V_{g_i,n}]_-\| < 1/j, \ \ \left\| (q_n h_n q_n - \theta q_n)_+ \right\| < 1/j \right\}$$

is again a finite intersection of sets in ω — the last constraint by (10) — hence lies in ω and in particular is infinite. Choose $n_1 < n_2 < \cdots$ with $n_j \in A_j$ and relabel along $(n_j)_j$. Then $q_n \neq 0$ for every n. For every fixed

$g \in \bar{H}$, the element $g = g_i$ occurs in the constraints defining A_j for all $j \geq i$, so $\|[q_n, V_{g,n}]_-\| \to 0$. The last constraint defining A_j also implies that the left-hand side of (10) tends to 0. Lemma 4.1 now gives an operator-norm asymptotic representation $(W_{g,n})$ of $\bar{H}$ on the corners $q_n M_{d_n}(\mathbb{C}) q_n$.

A 2-norm separation on the corner. Write $r_n = \operatorname{rank} q_n$ and equip the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with its renormalized trace tr_{r_n}. Applying tr_{r_n} to the ordinary form of (10) gives

$$\frac{1}{|S|} \sum_{a \in S} \operatorname{Re} \operatorname{tr}_{r_n}\big(q_n V_{a,n} q_n\big) \;=\; \operatorname{tr}_{r_n}\big(q_n h_n q_n\big) \;\leq\; \theta + o(1),$$

and polar correction moves each $q_n V_{a,n} q_n$ by $o(1)$ in operator norm, hence by $o(1)$ in tr_{r_n}, so the same bound holds with $W_{a,n}$ in its place: the average over S of the real parts of the renormalized traces of the compressed models is at most $\theta + o(1)$. For each n, at least one generator in S therefore has compressed model with real renormalized trace at most $\theta + o(1)$, and so normalized Hilbert–Schmidt distance at least $\sqrt{2(1-\theta)} - o(1)$ from the corner identity. Since S is finite, a subsequence realizes this at a single generator s_0.

The contradiction. The corner models form an operator-norm asymptotic representation of $\bar{H}$, and s_0 lies in $\bar{K} \leq \Theta(D)$. Hence s_0 is universally 2-norm trivial in the corner models: $\|W_{s_0,n} - 1\|_2 \to 0$, contradicting the positive lower bound above. $\square$

Theorem S2.7 (normal-Kazhdan obstruction)**.** *Let H be a countable group with a Kazhdan conjugation datum (Λ, ι, t, c), and let $K \trianglelefteq H$ be a normal subgroup with property* (T) *such that $K \subseteq N_{\mathrm{conj}}$. Then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(k) = 1$ for all $k \in K$.*

Since finite groups have property (T), Theorem S2.3 is the finite case of Theorem S2.7.

Proof. The first step of the proof of Theorem S2.3 shows that every element of N_{conj} is universally 2-norm trivial (Definition S2.2). Theorem S2.6 with $D = N_{\mathrm{conj}}$ gives the conclusion. $\square$

Every element of $\mathfrak{D}(H, L)$ is universally 2-norm trivial, so Theorem S2.6 gives the following normal-Kazhdan analogue of Theorem S2.5:

Corollary S2.8 (normal-Kazhdan form of the intrinsic residual)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $K \trianglelefteq H$ is a normal subgroup with property* (T) *and $K \leq \mathfrak{D}(H, L)$, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps every element of K to the identity.*

Proof. The proof of Theorem S2.5 shows that every element of $\mathfrak{D}(H, L)$ is universally 2-norm trivial in every operator-norm asymptotic representation of H; apply Theorem S2.6 with $D = \mathfrak{D}(H, L)$. $\square$

The subgroup generated by *all* normal property-(T) subgroups of $\mathfrak{D}(H, L)$ therefore also lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$.

S3. The finite-dimensional obstruction

Every finite-dimensional representation of E over any field maps w to the identity (Theorem 3.3 with $H = E$). Consequently finite-dimensional linear representations do not separate the points of E, and every finite quotient maps w to the identity. Since the Clifford quotient of Section 8 sends w to a nontrivial element, that quotient is infinite. The central involution w lies in the kernel of every homomorphism to a finite group, so E is a group of *Deligne type* in the terminology of [BDL, Definition 1.2]. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E is such a counterexample for the operator norm.

Theorem S3.1 (the exact intrinsic obstruction)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, and $\Gamma \leq H$ a subgroup. Then every homomorphism $\pi\colon H \to \mathrm{GL}(V)$ is trivial on the subgroup $\mathfrak{D}(H, \Gamma)$ of* (9).

Proof. Let C be the commutant of $\pi(\Gamma)$. The elements $g \in H$ for which conjugation by $\pi(g)$ preserves C in both directions form a subgroup. The commutant argument of Theorem 3.3 shows that every element of the compression semigroup P_Γ belongs to this subgroup, hence so does all of $G^+(\Gamma) = \langle P_\Gamma \rangle$. Thus, for $g \in G^+(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence every displayed generator in (9) lies in $\ker \pi$; since $\ker \pi \trianglelefteq H$, the whole normal closure $\mathfrak{D}(H, \Gamma)$ does as well. □

Corollary S3.2. *Every finite-dimensional linear representation of the group E of Definition 6.1, over any field, maps w to the identity, and every homomorphism from E to a finite group does as well. In particular E admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field.*

Proof. Apply Theorem 3.3 with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1; the hypotheses hold by the defining relations (8). Every finite-dimensional representation maps w to the identity, and $w \neq 1$ by Proposition 8.3. For a finite target, compose with its faithful left regular representation over $\mathbb{Q}$. The composite is a finite-dimensional linear representation and hence maps w to the identity; faithfulness of the left regular representation then implies that the original homomorphism also maps w to the identity. □

S4. The cyclic comparison

Fritz's marked $\mathrm{BS}(2,3)$ example is the context for this comparison [Fri, SV].

Theorem S4.1 (the cyclic comparison)**.** *Define the presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \;\middle|\; \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c,\gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}},\gamma_0] = [w_{\mathrm{BS}},t] = [w_{\mathrm{BS}},c] = 1 \end{array} \right\rangle,$$

where $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_{BS} is finitely presented and $w_{\mathrm{BS}} \neq 1$. There is an explicit surjective homomorphism from E_{BS} onto the subgroup of the Clifford semidirect product generated by the displayed images, and it maps w_{BS} to $\zeta \neq 1$. Nevertheless every finite-dimensional representation of E_{BS} over every field maps w_{BS} to the identity.

Proof of Theorem S4.1. The relator $w_{\mathrm{BS}}^2 = 1$ is redundant: tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (4). Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. By the commutant argument of Theorem 3.3, every finite-dimensional representation over every field maps w_{BS} to the identity.

For nontriviality, use the affine realization of $\mathrm{BS}(1,2)$ and its coset action. The base point is fixed by the cyclic base, while the two points represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these points and map c to the generator at the base point. All seven relators hold, and

$$w_{\mathrm{BS}} \longmapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\mathrm{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjection onto a Clifford quotient. □

Sharpness of the Kazhdan hypothesis. Running the Clifford representation of the preceding proof over finite site sets gives explicit unitary models of the seven relators in which w_{BS} is the scalar -1. Fix $m \geq 0$, put $M = 2m+3$ and $\zeta = e^{2\pi i/M}$, and take the coordinates $\mathbb{Z}/2 \times \mathbb{Z}/M$, the first for the Clifford sign and the second for the spectrum of the base. Let X and Z be the flip and the sign on the first coordinate, put $D = 2^{-1/2}(X+Z)$, a self-adjoint unitary, and set

$$\gamma_0(s,k) = (-1)^s \zeta^k, \qquad c = t^{-1}Dt,$$

where γ_0 is the displayed diagonal unitary, $b_0 = 0$, $b_1 = m+1$, and t is the pullback unitary

$$(t\xi)(s,k) = \xi(s, 2k + b_s), \qquad te_{(s,\,2k+b_s)} = e_{(s,k)}.$$

This is well defined because $(s,k) \mapsto (s, 2k+b_s)$ is a permutation when M is odd. Here γ_0 and t are monomial and c is an equal-weight sum of two monomial matrices. Since $tct^{-1} = D$ and $\gamma_0^2(s,k) = \zeta^{2k}$ is independent of the spin coordinate, the distinguished word is

$$w_{\mathrm{BS}} = [\,D,\; ZDZ\,] = (D \cdot ZDZ)^2 = -1,$$

because $ZDZ = 2^{-1/2}(Z - X)$ and $D \cdot ZDZ = \frac{1}{2}(XZ - ZX)$ squares to -1. Being a scalar, w_{BS} satisfies exactly the four relators $w_{\mathrm{BS}}^2 = 1$ and $[w_{\mathrm{BS}},\gamma_0] = [w_{\mathrm{BS}},t] = [w_{\mathrm{BS}},c] = 1$; with $c^2 = 1$ this is five of the seven.

The remaining two hold approximately. From $2(m+1) = M-1$ we get $\zeta^{m+1} = -e^{-i\pi/M}$, and

$$\left\|t\gamma_0 t^{-1} - \gamma_0^2\right\| = |1+\zeta^{m+1}| = 2\sin\frac{\pi}{2M}, \qquad \|[c,\gamma_0]-1\| \le 2|1+\zeta^{m+1}|;$$

D commutes with γ_0^2 exactly, so the lamp relator inherits the doubling defect and nothing else. Both are $O(1/m)$. Assembling the models over m gives $\Theta\colon E_{\mathrm{BS}} \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w_{\mathrm{BS}}) \neq 1$, and the image of Θ is MF because it embeds in $\mathcal{U}(\mathcal{Q})$. So the property (T) hypothesis in the central involution obstruction is essential: with the cyclic base $\langle\gamma_0\rangle$ in place of a Kazhdan base, some homomorphism to $\mathcal{U}(\mathcal{Q})$ is nontrivial on the distinguished word. Equivalently, in the residual language of Section S5, $w_{\mathrm{BS}} \notin \mathrm{Res}_{\mathrm{MF}}(E_{\mathrm{BS}})$.

The realized Clifford quotient is amenable: it is a subgroup of the semidirect product of the locally finite Clifford group on the coset space by the solvable affine group $\mathrm{BS}(1,2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$. It is therefore MF because the reduced C^*-algebra of every countable amenable group embeds into the universal UHF algebra [Sc20, Theorem B]. The models above give $\Theta(w_{\mathrm{BS}}) \neq 1$ for an MF image $\Theta(E_{\mathrm{BS}})$.

Exact finite-dimensional representations behave differently. In a finite image, the doubling relation gives $\pi(\langle\gamma_0^2\rangle) = \pi(\langle\gamma_0\rangle)$; since $\pi(tct^{-1})$ centralizes the first subgroup, it centralizes $\pi(\gamma_0)$, and the image of w_{BS} is the identity. To transfer this obstruction to exact local embeddings, take a finite set containing the generators together with every intermediate prefix of every defining word and of w_{BS}. Any exact local embedding of this set evaluates those words correctly and therefore satisfies the same equations. Hence neither E_{BS} nor its Clifford quotient is LEF, and the latter is not residually finite.

The scaling family. For $m \ge 2$, define E_m by replacing the three relations $tv_it^{-1} = v_i^2$ in the eight-generator presentation of E with $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged. Then $E_2 = E$. Write w_m for the image of the same distinguished word w.

Corollary S4.2 (the scaling family)**.** *For every natural number $m \ge 2$, the group E_m is finitely presented and w_m is a nontrivial central involution mapped to the identity by every homomorphism $E_m \to \mathcal{U}(\mathcal{Q})$. Consequently E_m is not MF, and neither $C^*_{\max}(E_m)$ nor $C^*_{\mathrm{r}}(E_m)$ is an MF C^*-algebra.*

Proof. Replace the affine dilation $\mathrm{diag}(2,2,2,1)$ by $\mathrm{diag}(m,m,m,1)$ (both acting as in Lemma 8.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Conjugating v_1 by the inverse dilation gives translation by e_1/m, which is not integral. Hence the two relevant cosets remain distinct, and the Clifford representation sends w_m to -1. The Kazhdan and central-involution arguments are unchanged for every $m \ge 2$. ∎

S5. The MF residual

Definition S5.1. For a countable group G_1, the *MF residual* is

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; \bigcap_{(d_n)} \;\; \bigcap_{\Theta\colon G_1\to\mathcal{U}(\mathcal{Q}((d_n)))} \ker\Theta,$$

a normal subgroup of G_1; the inner intersection is over all homomorphisms and the outer one over all sequences (d_n) of positive integers, in accordance with the convention fixed after (1).

If G_1/N is MF, compose $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$. The resulting homomorphism has kernel N, so $\mathrm{Res}_{\mathrm{MF}}(G_1) \leq N$. Thus the MF residual is contained in every normal subgroup whose quotient is MF. Proposition S5.3 shows that its own quotient is MF. It is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

Lemma S5.2 (functoriality)**.** *For every homomorphism $\varphi\colon G_1 \to G_2$ of countable groups,*

$$\varphi(\mathrm{Res}_{\mathrm{MF}}(G_1)) \;\subseteq\; \mathrm{Res}_{\mathrm{MF}}(G_2).$$

In particular, if $\varphi\colon E \to G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \mathrm{Res}_{\mathrm{MF}}(G_1)$ and Θ is a homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$, then $\Theta\circ\varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $\Theta(\varphi(x)) = 1$. For the second statement, Theorem A gives $w \in \mathrm{Res}_{\mathrm{MF}}(E)$. Hence $1 \neq \varphi(w) \in \mathrm{Res}_{\mathrm{MF}}(G_2)$. An injective homomorphism $G_2 \to \mathcal{U}(\mathcal{Q})$ would map $\varphi(w)$ to 1, a contradiction. ☐

Proposition S5.3 (the universal MF quotient)**.** *Let G_1 be a countable group and $R = \mathrm{Res}_{\mathrm{MF}}(G_1)$.*

1. *R is the kernel of a single homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$.*
2. *G_1/R is MF.*
3. *Every homomorphism from G_1 to an MF group factors through G_1/R.*

In particular G_1 is MF if and only if $R = \{1\}$.

Proof. (3) If $\varphi\colon G_1 \to M$ with M MF and $j\colon M \to \mathcal{U}(\mathcal{Q}((d_n)))$ is an injective homomorphism to $\mathcal{U}(\mathcal{Q})$, then $j\circ\varphi$ is a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$, so $j(\varphi(R)) = 1$ and, by injectivity of j, $\varphi(R) = 1$.

(1) and (2). Set $Q = G_1/R$. For every $\bar g \neq 1$ in Q, the definition of R gives a homomorphism $\Theta_{\bar g}\colon Q \to \mathcal{U}(\mathcal{Q})$ with $\Theta_{\bar g}(\bar g) \neq 1$. Enumerate the nonidentity elements $\bar g_1, \bar g_2, \dots$ and choose coordinate unitary lifts of the corresponding homomorphisms. At stage k, choose in the ith lift, for each $i \leq k$, a coordinate at which multiplicativity holds within $1/k$ on the first k group elements. Choose the same coordinate so that the designated element $\bar g_i$ remains separated by its fixed positive constant. Such coordinates exist because $\Theta_{\bar g_i}$ separates $\bar g_i$ along infinitely many coordinates. Take the block sum of the k chosen coordinates. Block operator norms are maxima, so the single block coming from $\Theta_{\bar g_i}$ already separates $\bar g_i$, and the sum therefore

separates all of $\bar{g}_1, \dots, \bar{g}_k$ at once. The resulting sequence is an operator-norm asymptotic representation separating every nonidentity element from the identity; its class, composed with the canonical unitary-corona isomorphism $\kappa_{(d_n)}$ of Lemma 2.2, is a faithful homomorphism $Q \to \mathcal{U}(\mathcal{Q})$. This proves (2), and composing with $G_1 \twoheadrightarrow Q$ proves (1). Finally, if G_1 is MF then an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ gives $R = \{1\}$; if $R = \{1\}$ then (2) says G_1 is MF. □

Corollary S5.4 (exact residual from a candidate quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$. Every homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$ factors uniquely through G_1/N. If in addition G_1/N is MF, then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) = N.$$

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every such Θ. For the reverse containment in the displayed equality, compose the quotient map $G_1 \to G_1/N$ with an injective homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$: its kernel is exactly N, while the MF residual lies in the kernel of every homomorphism to $\mathcal{U}(\mathcal{Q})$. □

Corollary S5.5 (residual reduction to the quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$ and let $q\colon G_1 \to G_1/N$ be the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(G_1) \;=\; q^{-1}(\mathrm{Res}_{\mathrm{MF}}(G_1/N)).$$

Proof. For the forward containment, Lemma S5.2 gives $q(\mathrm{Res}_{\mathrm{MF}}(G_1)) \subseteq \mathrm{Res}_{\mathrm{MF}}(G_1/N)$. Conversely, suppose $q(x) \in \mathrm{Res}_{\mathrm{MF}}(G_1/N)$ and let Θ be a homomorphism $G_1 \to \mathcal{U}(\mathcal{Q})$. Since $N \leq \mathrm{Res}_{\mathrm{MF}}(G_1)$, Θ factors as $\bar{\Theta} \circ q$ with $\bar{\Theta}$ a homomorphism $G_1/N \to \mathcal{U}(\mathcal{Q})$; hence $\Theta(x) = \bar{\Theta}(q(x)) = 1$. □

Corollary S5.6. *Let A be a unital C^*-algebra admitting an injective, possibly nonunital, $*$-homomorphism into a matrix quotient. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.*

Proof. Let $j\colon A \to \mathcal{Q}$ be the injective, possibly nonunital, $*$-homomorphism and let $\phi\colon E \to \mathcal{U}(A)$. Define

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

Then $\widehat{j}$ is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a homomorphism $E \to \mathcal{U}(\mathcal{Q})$. By Theorem A, $(\widehat{j} \circ \phi)(w) = 1$; injectivity of $\widehat{j}$ gives $\phi(w) = 1$. Proposition 8.3 has $w \neq 1$, so ϕ cannot be injective. □

S5.1. **Permanence and its failure.**

Lemma S5.7 (permanence)**.** *Let all groups be countable.*

(1) Every subgroup of an MF group is MF.
(2) Every residually finite group is MF.
(3) Every locally finite group is MF.

Proof. (1) Restrict an MF embedding of the ambient group to the subgroup. (2) For a finite test set F, choose a finite quotient that is injective on F and compose it with the left regular permutation action. This gives an exact finite-dimensional representation. Distinct permutations differ in a column of their permutation matrices and therefore have operator-norm distance at least 1. Proposition 2.3 now applies. (3) A finite test set F lies in a finite subgroup K. The left regular representation $\lambda_K\colon K \to \mathcal{U}(\mathbb{C}^K)$ is a finite-dimensional unitary representation of K by permutation matrices. Because a local model is defined on all of G, set

$$V(g) = \lambda_K(g) \quad (g \in K), \qquad V(g) = 1 \quad (g \notin K).$$

On the test set $F \subseteq K$, this map is exactly multiplicative, since products of test elements lie in K and λ_K is a homomorphism; the values outside K do not enter the two conditions. It separates as in (2). ☐

Corollary S5.8 (quotients)**.** *MF groups are not closed under quotients: the free group F_8 is MF, and the explicit group E is its quotient but is not MF.*

Proof. Map the eight free generators onto the eight displayed generators of E. The resulting homomorphism is surjective because those generators define the displayed presentation. The group F_8 is residually finite, hence MF by Lemma S5.7(2), while E is not MF by Theorem A. ☐

S6. Commuting orbits and the MF residual

In Theorem 4.2 the corner is the negative spectral corner of a central involution. Here it is an arbitrary projection of $\mathcal{Q}$ whose conjugation orbit commutes.

Theorem S6.1 (a commuting orbit forces commutation)**.** *Let H be countable, let $L \leq H$ have property* (T)*, let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Let $p \in \mathcal{Q}$ be a projection such that p commutes with $\Theta(s\gamma s^{-1})$ for every $\gamma \in L$ and the projections $\Theta(\gamma)\, p\, \Theta(\gamma)^*$, $\gamma \in L$, commute pairwise. Then p commutes with $\Theta(L)$.*

Proof. Property (T) makes L finitely generated; fix a finite symmetric generating set $S = \{a_1, \dots, a_m\}$. Write $p_g = \Theta(g)\, p\, \Theta(g)^*$ and $\Delta_g = p_g - p$ for $g \in L$, and suppose $\Delta_{g_0} \neq 0$ for some $g_0 \in L$.

Since p_g and p are commuting projections, Δ_g is self-adjoint with $\Delta_g^3 = \Delta_g$, so Δ_g^2 is a projection and $\Delta_g \Delta_g^2 = \Delta_g$; write $q_g = \Delta_g^2$. All the p_g commute, hence so do all the Δ_g and all the q_g. Put

$$q = 1 - \prod_{a \in S} (1 - q_a),$$

the join of the projections q_a, $a \in S$. The elements g with $p_g = p$ form a subgroup of L, so if $q = 0$, and hence $\Delta_a = 0$ for every $a \in S$, then $\Delta_g = 0$ for every $g \in L$, contradicting $\Delta_{g_0} \neq 0$. Thus $q \neq 0$.

The q-normalized Hilbert space. Take a self-adjoint bounded lift (x_n) of q. Since $\|x_n^2 - x_n\| \to 0$, for all sufficiently large n the spectrum of x_n avoids $[1/4, 3/4]$. Choose a continuous function $f\colon \mathbb{R} \to [0,1]$ with $f = 0$ on $(-\infty, 1/4]$ and $f = 1$ on $[3/4, \infty)$. Continuous functional calculus then gives, for all sufficiently large n, a projection $Q_n = f(x_n)$ with $\|Q_n - x_n\| \to 0$; define $Q_n = 0$ on the finitely many remaining coordinates. Thus (Q_n) is a projection lift of q.

Let $J = \{n : Q_n \neq 0\}$. Because $q \neq 0$, the set J is infinite. Restriction of coordinates induces a $*$-homomorphism

$$\rho_J\colon \prod_n M_{d_n}(\mathbb{C})/\bigoplus_n M_{d_n}(\mathbb{C}) \longrightarrow \prod_{n\in J} M_{d_n}(\mathbb{C})/\bigoplus_{n\in J} M_{d_n}(\mathbb{C}).$$

Replace Θ by $\rho_J \circ \Theta$ and every element introduced above by its image under ρ_J. All projection, commutation and cocycle identities are preserved, while the new image of q is still nonzero because its lift consists of nonzero projections and therefore has norm 1 at every coordinate of J. Relabel J by $\mathbb{N}$; then, without changing notation, we have $k_n = \operatorname{rank} Q_n \geq 1$ for every n.

Give $M_{d_n}(\mathbb{C})$ the inner product $\langle x, y\rangle_n = \operatorname{Tr}(y^*x)/k_n$, write $\|x\|_{\mathrm{F}} = \operatorname{Tr}(x^*x)^{1/2}$ for the unnormalized Hilbert–Schmidt norm, and let K_ω be the Hilbert-space ultraproduct of these spaces along a free ultrafilter ω.

The corner-supported ideal. Let $\mathcal{I}_q$ be the algebraic two-sided ideal generated by q, namely the set of finite sums $\sum_{j\leq r} x_j q y_j$ with $x_j, y_j \in \mathcal{Q}$. It is invariant under inner automorphisms. It is not closed in norm; the finite-sum description gives the rank bound (11) below.

For $z = \sum_{j\leq r} x_j q y_j \in \mathcal{I}_q$, choose bounded coordinate lifts $X_{j,n}$ of x_j and $Y_{j,n}$ of y_j and put $Z_n = \sum_{j\leq r} X_{j,n} Q_n Y_{j,n}$; then $\operatorname{rank}(Z_n) \leq r k_n$. The square of the Frobenius norm is the sum of the squares of the singular values; at most $\operatorname{rank}(Z_n)$ of them are nonzero and each is at most the operator norm, so

$$\|Z_n\|_{\mathrm{F}}^2 \;\leq\; \operatorname{rank}(Z_n)\,\|Z_n\|^2 \;\leq\; r\,k_n\,\|Z_n\|^2 \tag{11}$$

and (Z_n) is a bounded sequence for the inner products $\langle\cdot,\cdot\rangle_n$. Its class in K_ω depends only on $z = \sum_j x_j q y_j$: a sequence Z_n' obtained from a second decomposition, with r' summands, satisfies $\|Z_n - Z_n'\| \to 0$ and $\operatorname{rank}(Z_n - Z_n') \leq (r + r')k_n$, so $\|Z_n - Z_n'\|_{\mathrm{F}}^2/k_n \to 0$ by (11). Write $\Lambda(z) = [Z_n]_\omega$. The map $\Lambda\colon \mathcal{I}_q \to K_\omega$ is linear,

$$\|\Lambda(q)\| = \lim_\omega \Big(\frac{\operatorname{Tr} Q_n}{k_n}\Big)^{1/2} = 1,$$

and $\Lambda(z) = 0$ implies $\Lambda(xzy) = 0$ for all $x, y \in \mathcal{Q}$, since bounded lifts of x and y multiply $\|Z_n\|_{\mathrm{F}}$ by a bounded factor.

An exact cocycle. For $a \in S$ the projection q_a annihilates $\prod_{b\in S}(1 - q_b)$, so $q_a q = q_a$ and $\Delta_a = \Delta_a q_a = \Delta_a q \in \mathcal{I}_q$. From $p_{gh} = \Theta(g)\, p_h\, \Theta(g)^*$,

$$\Delta_{gh} = \Delta_g + \Theta(g)\,\Delta_h\,\Theta(g)^* \qquad (g, h \in L), \tag{12}$$

and $\mathcal{I}_q$ is inner-invariant, so $\Delta_g \in \mathcal{I}_q$ for every $g \in L$.

Choose unitary coordinate lifts $U_n(h)$ of $\Theta(h)$ (Lemma 2.2). The formula $\pi(h) = [\operatorname{Ad} U_n(h)]_\omega$ defines a unitary representation of H on K_ω: conjugation by a unitary preserves each $\langle \cdot, \cdot \rangle_n$, and $\|\operatorname{Ad} u - \operatorname{Ad} v\| \le 2\,\|u - v\|$. Moreover, $\pi(h)\Lambda(z) = \Lambda(\Theta(h)\, z\, \Theta(h)^*)$ on the closed invariant subspace $K_q = \overline{\Lambda(\mathcal{I}_q)}$. By (12) the map $\beta(g) = \Lambda(\Delta_g)$ satisfies

$$\beta(gh) = \beta(g) + \pi(g)\beta(h) \qquad (g, h \in L),$$

a 1-cocycle for $\pi|_L$ with values in K_q.

The cocycle is nonzero. Suppose $\beta(a) = 0$ for every $a \in S$. Put $r_1 = 1$ and $r_i = \prod_{j<i}(1 - q_{a_j})$ for $i \ge 2$. The products $e_i = r_i q_{a_i}$ are projections with $e_i = r_i - r_{i+1}$, so they are pairwise orthogonal and $\sum_i e_i = 1 - r_{m+1} = q$. Each Δ_{a_i} commutes with r_i, so $e_i = (\Delta_{a_i} r_i)\, \Delta_{a_i}$ and $\Lambda(e_i) = 0$, whence $\Lambda(q) = 0$ against $\|\Lambda(q)\| = 1$. So β is not identically zero.

Property (T). By the Delorme–Guichardet theorem [De, Gu], β is a coboundary: there is $y \in K_q$ with $\beta(g) = y - \pi(g)y$ for every $g \in L$. The hypothesis on p gives $\Delta_{s\gamma s^{-1}} = 0$, hence $\pi(s\gamma s^{-1})y = y$ for every $\gamma \in L$.

Normalize the Hilbert–Schmidt norms by $k_n = \operatorname{rank} Q_n$ rather than by d_n and apply Theorem S1.1 with that weight. In its proof, P is the projection onto $\operatorname{Fix} \pi(L)$ and the conjugate $Q = \pi(s) P \pi(s)^*$ onto $\operatorname{Fix} \pi(sLs^{-1})$, with $P \le Q$ and $P \sim Q$ in the same finite norm ultraproduct. Lemma 3.1 therefore gives $P = Q$. Hence $\pi(g)y = y$ and $\beta(g) = 0$ for every $g \in L$, contradicting the previous paragraph. Therefore $\Delta_g = 0$ for every $g \in L$ in the original corona; equivalently, p commutes with $\Theta(g)$ for every $g \in L$. □

Equivalently, every projection of $\Theta(sLs^{-1})' \cap \mathcal{Q}$ whose $\Theta(L)$-conjugates commute pairwise already lies in $\Theta(L)' \cap \mathcal{Q}$.

Definition S6.2 (involutive orbit witness)**.** Let H be a group, $L \le H$ a subgroup, and $s \in H$. An element $k \in H$ is an *involutive orbit witness* for (L, s) if

(W1) $k^2 = 1$;
(W2) $s\gamma s^{-1}$ commutes with k for every $\gamma \in L$;
(W3) the conjugates $\gamma_1 k \gamma_1^{-1}$ and $\gamma_2 k \gamma_2^{-1}$ commute for all $\gamma_1, \gamma_2 \in L$.

The pair (L, s) determines the normal subgroup

$$D_{\mathrm{coll}}(L, s) \;=\; \big\langle\!\big\langle\, [\gamma, k] \;:\; \gamma \in L,\ k \text{ a witness for } (L, s) \,\big\rangle\!\big\rangle_H.$$

In a permutational wreath product $K^{(X)} \rtimes G$ with $L \le G$ conjugated into itself by s and $x_0 \in X$ a point fixed by sLs^{-1}, every involution in the copy of K supported at x_0 is a witness: distinct coordinate copies commute, and sLs^{-1} fixes x_0.

Theorem S6.3 (the orbit of an involutive witness)**.** *Let H be countable, let $L \le H$ be a subgroup with property* (T)*, and let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$. Then*

$$D_{\mathrm{coll}}(L, s) \;\le\; \operatorname{Res}_{\mathrm{MF}}(H).$$

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and k a witness for (L, s). By (W1) the unitary $u = \Theta(k)$ is self-adjoint, so $p = \frac{1}{2}(1-u)$ is a projection, and

$$\Theta(\gamma)\, p\, \Theta(\gamma)^* = \tfrac{1}{2}\big(1 - \Theta(\gamma k \gamma^{-1})\big) \qquad (\gamma \in L).$$

By (W2) the projection p commutes with $\Theta(s\gamma s^{-1})$ for $\gamma \in L$, and by (W3) the displayed projections commute pairwise. By Theorem S6.1, p commutes with $\Theta(L)$; hence so does $u = 1 - 2p$, and $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. Kernels are normal, so Θ is trivial on $D_{\mathrm{coll}}(L, s)$. □

Remark S6.4 (finite-order witnesses). Condition (W1) can be relaxed to finite order. Let H be countable, let $L \le H$ have property (T), let $s \in H$ satisfy $s\gamma s^{-1} \in L$ for every $\gamma \in L$, and let $k \in H$ have finite order m. If sLs^{-1} centralizes k and the conjugates $\gamma k \gamma^{-1}$, $\gamma \in L$, commute pairwise, then every homomorphism $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta([\gamma, k]) = 1$ for every $\gamma \in L$. For $j \in \mathbf{Z}/m$, let

$$p_j = \frac{1}{m} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t/m}\, \Theta(k)^t.$$

These are the spectral projections of $\Theta(k)$. The hypotheses imply that $\Theta(sLs^{-1})$ centralizes every p_j and that their $\Theta(L)$-conjugates commute pairwise. Theorem S6.1 therefore makes every p_j commute with $\Theta(L)$. Fourier reconstruction, $\Theta(k) = \sum_j e^{2\pi i j/m} p_j$, then shows that $\Theta(k)$ also commutes with $\Theta(L)$. The normal closure of all such commutators, over all finite-order witnesses for (L, s), lies in $\mathrm{Res}_{\mathrm{MF}}(H)$, with equality whenever the corresponding quotient of H is MF.

Proof of Theorem B. Write $q\colon E \to E/\langle w \rangle$, $\bar{c} = q(\hat{c})$, and $\bar{L} = q(\iota(\mathcal{B}))$. The element $\bar{c}$ is an involution because $c^2 = 1$, and $q(t)\bar{L}q(t)^{-1}$ centralizes $\bar{c}$ because conjugating the relation $[c, \iota(\mathcal{B})] = 1$ of (8) by t gives $[\, t\iota(\mathcal{B})t^{-1},\, \hat{c}\,] = 1$ in E, where $\hat{c} = tct^{-1}$. The subgroup that centralizes $\hat{c}$ is the conjugate one, which is why the relation is conjugated by t. Property (T) passes to $\bar{L}$, and $q(t)$ conjugates it into itself. Thus (W1) and (W2) of Definition S6.2 hold, and only (W3) remains.

Pairwise commutation of the orbit. The linear generators x, y, z commute with $\hat{c}$, and conjugation by t sends each v_i to v_i^2, so $\iota(v_i^2)$ commutes with $\hat{c}$ as well. The conjugate $\iota(g)\, \hat{c}\, \iota(g)^{-1}$ therefore depends only on the exponent vector of the translation part of g modulo 2, so there are exactly eight such conjugates, indexed by $(\mathbb{Z}/2)^3$. By Theorem 5.1 the distinguished word is $w = [\hat{c}, \iota(v_1)\, \hat{c}\, \iota(v_1)^{-1}]$, the commutator attached to the class of v_1. The linear part acts through $\mathrm{SL}_3(\mathbb{Z}/2)$, which is transitive on the nonzero parity classes; the seven words $1, x, x^2, z, yx, y^2x, xyx$ in the linear generators send the class of v_1 to all seven nonzero classes, and each commutes with $\hat{c}$. Conjugating the displayed identity by the corresponding $\iota(r)$ shows that every nonzero parity commutator is a conjugate of w; the zero class gives $[\hat{c}, \hat{c}] = 1$. Since w is central, all eight lie in $\langle w \rangle$. Conjugating the commutator of two arbitrary conjugates by the inverse of one of them reduces it to this case. Hence all conjugates of $\bar{c}$ under $\bar{L}$ commute pairwise. Thus $\bar{c}$ is an involutive orbit witness for $(\bar{L}, q(t))$, and by Theorem S6.3 $[q(\iota(v_1)), \bar{c}] \in \mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$.

The homomorphism $E \to W$ of Proposition 8.3 sends $[\iota(v_1), \hat{c}]$ to the product of the Clifford generators at the two distinct points τo and $j(\bar{v}_1)\tau o$, and sends $\langle w\rangle$ into $\langle \zeta\rangle$. The product of the Clifford generators at two distinct points does not lie in $\langle \zeta\rangle$. Hence $[\iota(v_1), \hat{c}] \notin \langle w\rangle$, and its image in $E/\langle w\rangle$ is nontrivial. This image lies in the MF residual, so Proposition S5.3 implies that the quotient is non-MF. Corollary S5.5 identifies $\mathrm{Res}_{\mathrm{MF}}(E)$ with the q-preimage of $\mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle)$, so u belongs to it. □

Corollary S6.5 (collapse computes the residual)**.** *In the setting of Theorem S6.3, write $q\colon H \to H/D_{\mathrm{coll}}(L,s)$ for the quotient map. Then*

$$\mathrm{Res}_{\mathrm{MF}}(H) \;=\; q^{-1}\big(\mathrm{Res}_{\mathrm{MF}}(H/D_{\mathrm{coll}}(L,s))\big),$$

and if $H/D_{\mathrm{coll}}(L,s)$ is MF, then $\mathrm{Res}_{\mathrm{MF}}(H) = D_{\mathrm{coll}}(L,s)$*.*

Proof. Combine Theorem S6.3 with Corollaries S5.5 and S5.4. □

If $H/D_{\mathrm{coll}}(L,s)$ is MF, then it is the universal MF quotient of H (Proposition S5.3): every homomorphism from H to an MF group factors through it.

S6.1. **The exact residual of E.** For the literal group E, put $d = \hat{c} = tct^{-1}$, $u = [d, v_1]$, and $D = \langle\!\langle u\rangle\!\rangle_E$. Then

$$\mathrm{Res}_{\mathrm{MF}}(E) = D = N_{\mathrm{conj}}, \qquad E/D \cong \Sigma *_{\mathcal{B}} (\mathcal{B} \times C_2),$$

and E/D is MF. Moreover, if $q\colon E \to E/\langle w\rangle$ and D_{coll} is the collapse defect used in the proof of Theorem B, then

$$\mathrm{Res}_{\mathrm{MF}}(E/\langle w\rangle) = D_{\mathrm{coll}} = D/\langle w\rangle, \qquad \mathrm{Res}_{\mathrm{MF}}(E) = q^{-1}(D_{\mathrm{coll}}).$$

Proof. d commutes with the linear part of $\mathcal{B}$ and with every even translation. Hence $[d,b]$ depends only on the translation part of b modulo 2. The linear part is transitive on the seven nonzero classes of $(\mathbb{Z}/2)^3$, by the explicit seven-word calculation used in the proof of Theorem B; every nontrivial $[d,b]$ is therefore a conjugate of u. This proves $D = N_{\mathrm{conj}}$. Moreover, $u \in \mathrm{Res}_{\mathrm{MF}}(E)$ by Theorem B; since the residual is normal, $D = \langle\!\langle u\rangle\!\rangle_E \le \mathrm{Res}_{\mathrm{MF}}(E)$. Universal 2-norm triviality alone does not imply this corona statement (Section S9).

In the quotient by D, the involution d commutes with all of $\mathcal{B}$. Replacing c by d, so that $c = t^{-1}dt$, makes the old relation $[c,\mathcal{B}] = 1$ redundant after conjugation by t; the central-mark relations are redundant as well because $w = u^2$. The remaining presentation is precisely

$$\langle \mathcal{B}, t, d : \; tbt^{-1} = \alpha(b), \; d^2 = 1, [d,\mathcal{B}] = 1\rangle \cong \Sigma *_{\mathcal{B}} (\mathcal{B} \times C_2).$$

The group Σ is finitely generated linear in characteristic zero and hence residually finite [Mal]. Put $G_0 = \Sigma \times C_2$. Choose a separating chain of finite quotients of G_0, let A be the C^*-algebra generated by their regular representations in the norm matrix corona, and let τ be an ultralimit of the normalized traces. The algebra A is separable because it is generated by the countable family of group unitaries; it is MF and $\tau(g) = 0$ for every $g \neq 1$. Put $C = C^*(\mathcal{B}) \subseteq A$. Shulman's symmetric amalgam theorem [Sh, Theorem 10] makes the full unital amalgam $A *_C A$ MF. The canonical homomorphism $G_0 *_{\mathcal{B}} G_0 \to \mathcal{U}(A *_C A)$ is injective. Indeed, in the GNS von Neumann algebra of (A, τ), the trace-preserving conditional expectation onto C'' sends every group unitary outside $\mathcal{B}$ to zero. The reduced amalgam trace therefore vanishes on every nontrivial Bass–Serre reduced word. Hence $G_0 *_{\mathcal{B}} G_0$ is MF. Normal forms for amalgamated free products give an embedding

$$\Sigma *_{\mathcal{B}} (\mathcal{B} \times C_2) \hookrightarrow G_0 *_{\mathcal{B}} G_0,$$

using the first vertex group for Σ and the second for $\mathcal{B} \times C_2$. Subgroups of MF groups are MF, so E/D is MF. Corollary S5.4 now gives $\mathrm{Res}_{\mathrm{MF}}(E) = D$. Applying Corollary S5.5 first to $\langle w \rangle \le D$ and then to the collapse defect gives the two identities downstairs. Explicitly, $q(u)^{-1} = [q(\iota(v_1)), \bar{c}]$ is one of the generators of D_{coll}, so $D/\langle w \rangle \le D_{\mathrm{coll}}$; the reverse inclusion follows from the residual calculation just proved. $\square$

S7. Group algebras, soficity, and permanence

S7.1. The reduced group algebra of E.

Lemma S7.1. *Let G_1 be a countable discrete group.*

(1) The canonical tracial state τ on $C^*_{\mathrm{r}}(G_1)$ is faithful.
(2) Every unital C^*-algebra A with a faithful tracial state is stably finite: for every $k \ge 1$, every isometry in $M_k(A)$ is a unitary.

Proof. (1) Realize $C^*_{\mathrm{r}}(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g. Let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$. Each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C^*_{\mathrm{r}}(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}} x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k}\sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k}\sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$. It is faithful. Indeed, the i-th diagonal entry of x^*x is $\sum_j x_{ji}^* x_{ji}$. If $\tau_k(x^*x) = 0$, then $\tau(x_{ji}^* x_{ji}) = 0$ and hence $x_{ji} = 0$ for all i, j. If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

S7.2. The Clifford quotient and permanence. The map $E \to W$ of Proposition 8.3 is surjective: its image contains Σ and the generator c_o at the base point, hence every Clifford generator by transitivity of the Σ-action on X. Since E is generated by v_1, x, y, z, t, c (Section 7), their six images generate W. The map sends $w \in \mathrm{Res}_{\mathrm{MF}}(E)$ to the nontrivial sign ζ, so $\zeta \in \mathrm{Res}_{\mathrm{MF}}(W)$ by Lemma S5.2. Hence the finitely generated group W is not MF. Neither is $C^*_{\mathrm{r}}(W)$: if $C^*_{\mathrm{r}}(W)$ embedded in a matrix quotient, the map $u \mapsto e(u) + 1 - e(1)$ would give an injective homomorphism $W \to \mathcal{U}(\mathcal{Q})$, as in the proof of Theorem A.

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \mathrm{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel is locally finite — hence LEF, sofic and MF — and whose quotient is MF, while W itself is finitely generated and not MF.

Proof. The Clifford group $\mathrm{Cl}(X)$ is locally finite: every finitely generated subgroup involves only finitely many sites, hence is contained in the corresponding finite Clifford group. Therefore $\mathrm{Cl}(X)$ is MF by Lemma S5.7(3). The group Σ identifies with the subgroup of $\mathrm{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D: the levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant gives the $\mathbb{Z}$-coordinate. It is residually finite. Modulo every odd integer, the dilation matrix is invertible because its determinant is 8; hence the congruence reduction of the integral affine model is compatible with the dilation and extends over the direct limit and its shift. These finite quotients separate every nontrivial element of Σ, consistent with Mal'cev's theorem for finitely generated linear groups [Mal]. Thus Σ is MF by Lemma S5.7(2), whereas W is not MF by the preceding construction. ☐

Extensions by $\mathbb{Z}$. The two canonical projections define a surjection $W \to \mathbb{Z}$ with kernel $\mathrm{Cl}(X) \rtimes T$. Every finite subset of the kernel lies in $L \rtimes \bar{\Gamma}_n$ for some finite invariant subgroup $L \leq \mathrm{Cl}(X)$ and some level $\bar{\Gamma}_n$. This semidirect product is residually finite because the action on L factors through a finite quotient. The kernel is therefore LEF and hence MF. Thus W is a non-MF extension of an LEF group by $\mathbb{Z}$. It is nevertheless sofic by the Elek–Szabó amenable-extension theorem [ES06].

A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\mathrm{Res}_{\mathrm{MF}}(S_{\mathrm{simp}}) = S_{\mathrm{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed E, which is sofic by Theorem D, in such a group. By functoriality of the MF residual, the image of w is a nontrivial element of the MF residual of the simple group. This residual is a nontrivial normal subgroup, so simplicity makes it the whole group. Every homomorphism to an MF group maps the MF residual to 1. ☐

S7.3. **Soficity of E.** Deleting the letter c and the families (C) and (W) from (8) leaves the presentation $\langle\, \mathcal{B}, t \mid tut^{-1} = \bar{\alpha}(u) \,\rangle$ of the ascending HNN extension of the base along the doubling endomorphism. By Remark 7.2 the base is $\bar{\Gamma}$, so

$$E/\langle\!\langle c\rangle\!\rangle \;\cong\; \Sigma$$

for the group $\Sigma = T \rtimes \mathbb{Z}$ of Section 8, with its site set $X = \Sigma/B$, base point o, shift τ and level-zero base $B = j(\bar{\Gamma})$. The levels

$$\bar{\Gamma}_n = \tau^{-n} B \tau^n \qquad (n \geq 0)$$

form an ascending chain of copies of $\bar{\Gamma}$ with union T. Put $K = \tau^{-1}B\tau$; then $B \leq K$ and $[K : B] = [\bar{\Gamma} : \bar{\alpha}(\bar{\Gamma})] = 8$. The cosets Σ/K are the *blocks*, and each block, as a set of sites, has $[K : B]$ elements.

Let $\mathcal{G}$ be the graph on X whose edges are the Σ-orbit of the pair $\{\tau o,\, j(\bar{v}_1)\tau o\}$, whose entries are distinct by Lemma 8.2. Two sites are adjacent in $\mathcal{G}$ exactly when they are distinct and lie in a common block. Adjacent sites share a block because the two sites are gB and hB for the representatives $g = \tau$ and $h = j(\bar{v}_1)\tau$, and $g^{-1}h = \tau^{-1}j(\bar{v}_1)\tau \in K$ since $j(\bar{v}_1) \in B$ and $K = \tau^{-1}B\tau$; hence $gK = hK$. (Sites are cosets, so the product and inverse have to be taken in Σ on representatives, not on the sites themselves.) Conversely, the bijection $u\,\bar{\alpha}(\bar{\Gamma}) \mapsto \tau^{-1}j(u)\tau o$ identifies the block of o with $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ and the generating edge with the pair $\{\bar{\alpha}(\bar{\Gamma}),\, \bar{v}_1\bar{\alpha}(\bar{\Gamma})\}$. Left multiplication by $\bar{\alpha}(\bar{\Gamma})$ on the coset space $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ is transitive on the nontrivial cosets, so every pair of distinct sites of a block is the image of the generating pair. Let $L = \langle x, y, z\rangle$ be the linear part, so $\bar{\Gamma} = \mathbb{Z}^3 \rtimes L$. Since $\bar{\alpha}$ squares the three translations and fixes x, y, z (Lemma 8.2), its image is $(2\mathbb{Z}^3) \rtimes L$. Reduction of the translation coordinate modulo 2 identifies $\bar{\Gamma}/\bar{\alpha}(\bar{\Gamma})$ with $(\mathbb{Z}/2)^3$. Under that identification left multiplication by an element of $\bar{\alpha}(\bar{\Gamma})$ with linear part N and (necessarily even) translation part sends a class $\bar{u}$ to $\bar{N}\bar{u}$, so the action factors through the linear action of the reduction $\bar{L} \leq \mathrm{GL}_3(\mathbb{F}_2)$ of L. The seven words

$$1, \quad x, \quad x^2, \quad z, \quad yx, \quad y^2x, \quad xyx$$

send $\bar{e}_1$ to $\bar{e}_1$, $\bar{e}_3$, $\bar{e}_2$, $(0, 1, 1)$, $(1, 1, 0)$, $(1, 1, 1)$, $(1, 0, 1)$, the seven nonzero classes. This proves the required transitivity.

For a graph $\mathcal{G}$ on X, let $C(\mathcal{G})$ be the group presented by generators ζ and c_ξ ($\xi \in X$) and relations

$$\zeta^2 = c_\xi^2 = 1, \qquad [\zeta, c_\xi] = 1, \qquad [c_\xi, c_\eta] = \zeta \;\text{ for each edge } \{\xi, \eta\} \text{ of } \mathcal{G}.$$

For the complete graph on X this is $\mathrm{Cl}(X)$ of Construction 8.1, and adjoining the anticommutation relations at the remaining pairs gives a surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$.

Proposition S7.2 (normal form for E)**.** *There is an isomorphism*

$$E \;\cong\; C(\mathcal{G}) \rtimes \Sigma$$

under which $\langle\!\langle c\rangle\!\rangle$ is the factor $C(\mathcal{G})$, the conjugate gcg^{-1} is the lamp c_{gB}, and the distinguished word w is the sign ζ. Each level $\bar{\Gamma}_n$ has finite orbits on X.

Proof. Send the six base letters and t to the corresponding generators of Σ, and c to c_o. Families (B) and (T) of (8) hold in Σ by construction. For family (C): $c_o^2 = 1$, and B fixes o, so each base generator fixes c_o. For family (W): the pair $\{\tau o,\, j(\bar{v}_1)\tau o\}$ is an edge, so the image of w is $[c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = \zeta$, which is central in $C(\mathcal{G})$ and fixed by Σ. This defines $E \to C(\mathcal{G}) \rtimes \Sigma$.

In the other direction, send ζ to w and c_{gB} to gcg^{-1}. The assignment is independent of the coset representative, because the base centralizes c. The images are involutions, and w is central of order two. Along an edge $g\{\tau o,\, j(\bar{v}_1)\tau o\}$ the commutator of the two images is $gwg^{-1} = w$, and $C(\mathcal{G})$ imposes relations only along edges. So the assignment respects the defining relations of $C(\mathcal{G})$. Combined with the splitting $\Sigma \to E$ on the seven remaining letters, this gives $C(\mathcal{G}) \rtimes \Sigma \to E$. The two composites fix the eight letters of E and the generators ζ, c_ξ of $C(\mathcal{G})$, hence are the identity.

For the orbits, $\bar{\alpha}(\bar{\Gamma})$ has index eight in $\bar{\Gamma}$ — the parity classes of the translation lattice — so τ conjugates B onto a subgroup of finite index in itself. Every $\bar{\Gamma}_n$ is therefore commensurable with B, all of Σ commensurates B, and orbit–stabilizer makes each $\bar{\Gamma}_n$-orbit in $X = \Sigma/B$ finite. $\square$

Composing the isomorphism with $C(\mathcal{G}) \to \mathrm{Cl}(X)$ gives the surjection $E \to W$ of Proposition 8.3. The group $\mathrm{Cl}(X)$ is locally finite; $C(\mathcal{G})$ is not, since two lamps at sites of different blocks generate an infinite dihedral group and three blocks give a nonabelian free subgroup. For a finite set S of sites write

$$M_S = \langle\, \zeta,\ c_\xi\ (\xi \in S)\,\rangle \ \le\ C(\mathcal{G}).$$

This distinction prevents a misleading wreath-product identification. The auxiliary quotient $W/\langle\zeta\rangle$ is an ordinary permutational lamp group, but $C(\mathcal{G})/\langle\zeta\rangle$ is the free product of the elementary abelian 2-groups attached to the blocks of $\mathcal{G}$. Thus the present normal form does not identify $E/\langle w\rangle$ with an ordinary permutational wreath product. It is nevertheless closely related to central extensions of lamplighter groups [FFG, Bo25], to the generalized wreath products of Kun–Thom [KT26], to the sofic-action framework of Gao, Kunnawalkam Elayavalli, and Patchell [GKEP], and to the halo-product framework of Alekseev–Bradford [AB].

Lemma S7.3 (finite-site subgroups are residually finite)**.** *For every finite set S of sites the subgroup M_S is residually finite.*

Proof. The quotient map by $\langle\zeta\rangle$ maps $C(\mathcal{G})$ onto the free product of the elementary abelian 2-groups of the blocks, and M_S into the free product of the finitely many factors at blocks meeting S. A free product of finite groups is residually finite. The kernel of $C(\mathcal{G}) \to C(\mathcal{G})/\langle\langle\zeta\rangle\rangle$ is the normal closure of ζ, and ζ is central, so that normal closure is $\langle\zeta\rangle$; hence the kernel of this map on M_S lies in $\langle\zeta\rangle$. The surjection $C(\mathcal{G}) \to \mathrm{Cl}(X)$ sends ζ to the sign of $\mathrm{Cl}(X)$, which is nontrivial, and $\mathrm{Cl}(X)$ is locally finite, so the image of the finitely generated subgroup M_S in $\mathrm{Cl}(X)$ is finite. An element of M_S in the kernel of the first map lies in $\langle\zeta\rangle$, and the second map separates the two elements of that subgroup. $\square$

Proof of Theorem D. Let E_0 be the kernel of the homomorphism $E \to \mathbb{Z}$ given by the exponent sum in t. By Proposition S7.2, $E_0 \cong C(\mathcal{G}) \rtimes T$.

Take a finite subset of E_0. Its T-coordinates lie in one level $\bar{\Gamma}_n$, the chain being ascending, and its $C(\mathcal{G})$-coordinates involve only finitely many lamp generators and possibly the sign, so they lie in M_{S_0} for some finite set S_0 of sites. Let S be the union of the $\bar{\Gamma}_n$-orbits of the sites in S_0, a finite set by Proposition S7.2. Then M_S is $\bar{\Gamma}_n$-invariant and contains M_{S_0}, so the finite subset lies in $M_S \rtimes \bar{\Gamma}_n$.

That group is residually finite. Indeed M_S is residually finite by Lemma S7.3, the level $\bar{\Gamma}_n \cong \bar{\Gamma}$ is residually finite through the congruence quotients of its matrix model (Lemma 8.2), and the action of $\bar{\Gamma}_n$ on M_S factors through a finite permutation group F because S is finite. The kernel of $\bar{\Gamma}_n \to F$ has finite index, and the corresponding finite-index subgroup of the semidirect product is $M_S \times \ker(\bar{\Gamma}_n \to F)$, which is residually finite; residual finiteness passes to a group containing a residually finite subgroup of finite index.

Hence E_0 is a directed union of subgroups that embed in residually finite groups; that is, E_0 is LEF and therefore sofic. The group E is a split extension of E_0 by $\mathbb{Z}$, split by t, so the Elek–Szabó amenable-extension theorem implies that E is sofic [ES06], and hence hyperlinear. By Theorem A, E is not MF. LEF groups are MF [CDE], so E is not LEF; a residually finite group is LEF, so E is not residually finite either. $\square$

Exactness. The same decomposition also proves that E is exact. The quotient $C(\mathcal{G})/\langle\zeta\rangle$ is the free product of the elementary abelian 2-groups attached to the blocks, each of order 2^8. There are countably many blocks; enumerate them, and let Q_n be the free product of the factors attached to the first n. Each factor is finite, hence exact, so Q_n is a free product of finitely many exact groups and is exact by Dykema's theorem [Dy]. The Q_n increase with union the whole quotient. In a discrete group every subgroup is open, so a group that is the union of an increasing family of exact subgroups is exact [KW1, Lemma 2.5], and the quotient is exact. Since $\langle\zeta\rangle$ is finite, extension permanence [KW2] gives exactness of $C(\mathcal{G})$. The group $\Sigma \leq \mathrm{GL}_4(\mathbb{Q})$ is exact by Guentner–Higson–Weinberger [GHW]. Finally $E \cong C(\mathcal{G}) \rtimes \Sigma$ is an extension, hence is exact by [KW2]. For a discrete group, group exactness is equivalent to exactness of its reduced group C^*-algebra [KW1]; therefore $C^*_{\mathrm{r}}(E)$ is exact.

S8. Quasi-identities and undecidability

S8.1. **A finite universal Horn obstruction.** The presentation of E yields a finite quasi-identity — equivalently, a universal Horn sentence — satisfied by every MF group. Write R for the forty-one relators of (8), regarded as words in eight variables, and w for the distinguished word in those variables.

Proposition S8.1 (finite quasi-identity)**.** *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1,\dots,x_8\colon \quad \bigwedge_{r\in R} r(x_1,\dots,x_8)=1 \implies w(x_1,\dots,x_8)=1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w\neq 1$.

Proof. A tuple satisfying the relators induces a homomorphism $E\to G$ sending the distinguished word of E to $w(x_1,\dots,x_8)$. If G is MF, the image is trivial, because $w\in \mathrm{Res}_{\mathrm{MF}}(E)$ by Theorem A and the residual is functorial (Lemma S5.2). The canonical tuple of E satisfies R by definition, and its distinguished word is nontrivial by Theorem A. □

In the space of eight-generator marked groups, the groups satisfying all relators of R and $w\neq 1$ form a nonempty clopen set. Both conditions depend on finitely many word equalities and inequalities, and E belongs to this set. Proposition S8.1 excludes every MF group from it.

S8.2. **Undecidability of MF recognition.** Lemma S5.7(1) and Theorem A show that MF is a Markov property of finitely presented groups: the trivial group is MF, while by subgroup heredity the finitely presented group E embeds in no MF group. The Adian–Rabin theorem therefore implies that MF recognition is undecidable [Ra].

Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_e for the group presented by the code e.

Corollary S8.2 (undecidability of MF recognition)**.** *Write W for the predicate on pairs consisting of a presentation code and a word in the generators of that code, holding when the word is trivial in the presented group. Then:*

(1) W is undecidable;
(2) no algorithm decides, from a presentation code, whether the presented group is MF;
(3) W is recursively enumerable, whereas its complement and the set of presentation codes of non-MF groups are not.

Proof. For the undecidability of W, use Boone's finitely presented group with undecidable word problem [Bo]. A decision procedure for the uniform predicate W would decide the word problem of this one fixed presentation, which is impossible.

For the reduction we use the effective form of the Adian–Rabin construction, which supplies a computable map Δ from pairs (a presentation code e, a word u in its generators) to presentation codes such that

- if $u = 1$ in G_e, then $\Delta(e, u)$ presents an MF group: the trivial group in the classical form, and a free group in the variant used here, a free group being residually finite and so MF by Lemma S5.7(2); and
- if $u \neq 1$ in G_e, then G_e embeds in the group presented by $\Delta(e, u)$.

This is Rabin's theorem [Ra] and, in the form with the explicit computable output presentation, Adian's [Ad]; an English translation of Adian's papers with the construction written out is [NB]. Apply Δ to the free product of the input presentation with a fixed presentation of E and to the input word. The resulting presentation code is computable from the pair and presents an MF group exactly when the input word is trivial. In the trivial case the output group is free and hence MF. In the nontrivial case the free product, and therefore E, embeds in the output group. The output group is then non-MF by Lemma S5.7(1). Composing a decision procedure for MF recognition with this map would therefore decide W, so no such procedure exists. For the last assertion, the positive instances of W are exactly those carrying a finite certificate — a product of conjugates of relators freely equal to the given word — and whether a given certificate works is decidable, so W is recursively enumerable. Were its complement recursively enumerable as well, W would be decidable, which it is not. Finally, the preimage of the set of non-MF presentation codes under this computable map is exactly the set of pairs outside W, and the preimage of a recursively enumerable set under a computable map is recursively enumerable. □

Multiplicity. There are infinitely many pairwise nonisomorphic finitely presented non-MF groups, and continuum many pairwise nonisomorphic finitely generated non-MF groups. The groups $E\times\mathbb{Z}^k$ for $k \geq 0$ are finitely presented, pairwise nonisomorphic, and non-MF, each containing E (Lemma S5.7(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks. For finitely generated examples, choose continuum many pairwise nonisomorphic finitely generated groups N, for instance Neumann's two-generator groups [Ne]. Each product $E\times N$ is non-MF. A fixed countable group X can be isomorphic to $E\times N$ for at most countably many isomorphism types of N, because each such N embeds in X as a finitely generated subgroup and X has only countably many finitely generated subgroups. Hence the products yield $2^{\aleph_0}$ pairwise nonisomorphic finitely generated non-MF groups.

S9. Limitations of the operator-norm method

The argument requires operator-norm control. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. The estimate $\|\mathrm{Ad}(U) - \mathrm{Ad}(V)\| \leq 2\,\|U - V\|$ has no converse: unitaries can be arbitrarily close in normalized Hilbert–Schmidt norm while their adjoint operators remain at operator-norm distance 2. Operator-norm control ensures that the classes $[\operatorname{Ad} U_n(g)]_\omega$ are multiplicative; Hilbert–Schmidt control alone does not provide this.

Adding an identity block leaves operator-norm multiplicative defects and separations unchanged but decreases normalized Hilbert–Schmidt distances. By taking the identity block sufficiently large, all pairwise normalized Hilbert–Schmidt distances can be made smaller than any prescribed positive bound. Thus universal 2-norm triviality of an element does not by itself prove a group non-MF. The central-involution argument passes to an invariant corner and equips it with the trace normalized by its own rank r_n: the 2-norm separation on the corner is measured against r_n rather than d_n, so r_n/d_n may tend to zero without affecting the contradiction. The spectral projection of Theorem S2.7 is normalized on its own range in the same way, and the proof of Theorem S6.1 normalizes by the rank k_n.

Stable finiteness and faithful traces on the coordinate algebras do not suffice for Theorem 3.5. For a finite factor M in its standard representation on $L^2(M,\tau)$, the right action is the commutant of the left action. Hence the von Neumann algebra generated by the two actions is $(M \cup M')'' = Z(M)' = B(L^2(M,\tau))$; in infinite dimension this is not a finite C^*-algebra. Constant coordinates $A_n = C^*_{\mathrm{r}}(E)$ give an explicit counterexample: the left regular representation embeds E exactly and injectively into the norm quotient

$$\prod_n A_n / \bigoplus_n A_n,$$

the element c centralizes the Kazhdan base there, and conjugation by t produces tct^{-1}, whose commutator with the base is the element u with $u^2 = w \neq 1$ of Theorem A. Hence Theorem 3.5 fails when its matrix coordinates are replaced by stably finite unital C^*-algebras carrying faithful tracial states and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Their question concerns Hamming and tracial models, whereas the adjoint spectral estimates here require operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply.

Unitary lifting over general C^*-algebras. If u is unitary in $\prod_n A_n / \bigoplus_n A_n$ for unital C^*-algebras A_n, and (x_n) is any bounded lift, then $\|x_n^* x_n - 1\| \to 0$ and $\|x_n x_n^* - 1\| \to 0$. Thus x_n is invertible for all large n, and the polar correction $u_n = x_n(x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \to 0$. This argument uses only polar correction in the coordinate algebras and requires neither real rank zero nor semiprojectivity in the sense of Loring [Lo98]. Theorem 3.5 additionally uses the Hilbert-space conjugation model and finiteness of the norm ultraproduct.

S10. A torsion-free finitely presented non-MF group

Theorem S10.1 (a torsion-free example)**.** *There exists a torsion-free finitely presented non-MF group.*

Proof. The construction in [FFF, Section 2] starts from a torsion-free finitely presented double HNN extension and applies Hull's common-quotient theorem to obtain a torsion-free finitely presented acylindrically hyperbolic property-(T) quotient G_0. In the notation of that construction, G_0 contains a Kazhdan subgroup Γ, a stable letter t_1, and an embedded finitely presented infinite simple torsion-free group $S' \leq \Gamma$ such that

$$t_1 \Gamma t_1^{-1} \subseteq \Gamma, \qquad t_1^{-1} S' t_1 \subseteq C_{G_0}(\Gamma).$$

For $s \in S'$, put $z = t_1^{-1} s t_1$. Then $z \in C_{G_0}(\Gamma)$ and $t_1 \in P_\Gamma \subseteq G^+(\Gamma)$, so for every $\ell \in \Gamma$,

$$[s, \ell] = [t_1 z t_1^{-1}, \ell] \in \mathfrak{D}(G_0, \Gamma).$$

Thus $[S', S'] \leq \mathfrak{D}(G_0, \Gamma)$. The infinite nonabelian simple group S' is perfect, and hence $S' \leq \mathfrak{D}(G_0, \Gamma)$.

Put $N = \langle\!\langle S' \rangle\!\rangle_{G_0}$. This subgroup is infinite and normal, hence s-normal. By Osin's lemma, N acts non-elementarily [Os, Lemma 7.1]. The finite radical of the torsion-free group G_0 is trivial, so N is suitable. Apply Theorem 7.1 of [Hu] with the elements t_i equal to a finite generating set of G_0 and with an injectivity radius containing a fixed nonidentity element of S', and write $q \colon G_0 \twoheadrightarrow G$ for the resulting quotient. The group G is torsion-free by Theorem 7.1(e), finitely presented because the construction adds finitely many relators, and property-(T) because it is a quotient of G_0. Condition (c) gives that the image of N equals G. The image of S' is nontrivial, and its kernel is a proper normal subgroup of the simple group S', so $q|_{S'}$ is injective. It follows that $q(S')$ normally generates G.

Put $L = q(\Gamma)$. The relations above descend to

$$q(t_1) L q(t_1)^{-1} \subseteq L, \qquad q(t_1)^{-1} q(S') q(t_1) \subseteq C_G(L).$$

The same commutator calculation gives $q(S') \leq \mathfrak{D}(G, L)$. Since $q(S')$ normally generates G and $\mathfrak{D}(G, L)$ is normal, $\mathfrak{D}(G, L) = G$. Applying Corollary S2.8 with $K = G$ shows that every homomorphism $G \to \mathcal{U}(\mathcal{Q})$ is trivial. The embedded copy of S' makes G nontrivial, so no such homomorphism is injective and G is not MF.

The normal-Kazhdan criterion is essential here because a torsion-free group has no nontrivial finite subgroup. $\square$

Remark S10.2 (a small-cancellation realization). The affine base of Sections 7–8 can be replaced by the small-cancellation construction of [FFF, §2], which Fournier-Facio pointed out to us. There a universal finitely presented torsion-free group — one containing a copy of every finitely presented torsion-free group — is embedded by small cancellation in a finitely presented torsion-free property-(T) group P. Hence that P contains every finitely presented torsion-free group, in particular a direct product $P_1 \times P_2$

with $P_i \cong P$; loc. cit. states this as $P_1 \times P_2 \times S \leq P$ with S finitely presented simple, and the third factor is not needed here. Choosing an isomorphism $\alpha\colon P \to P_1 \leq P$ and any $a \in P_2 \setminus \{1\}$ gives the hypotheses of Theorem 5.1. The group $E(P, \alpha, a)$ has torsion even though P is torsion-free, since the central involution w of Theorem 5.1 is nontrivial.

S11. Open questions

The construction leaves the following questions open:

(1) Does MF imply hyperlinearity? An MF embedding controls operator norm but does not require the induced tracial model to separate nontrivial group elements, so it does not directly produce a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.

(2) Does the obstruction survive when the matrix blocks are replaced by other coordinate algebras? The proof of Theorem 3.5 uses finite dimensionality of $M_{d_n}(\mathbb{C})$ for the conjugation action on $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and for finiteness of the norm ultraproduct. These conclusions need not hold for arbitrary faithfully traced coordinate algebras. For which separable C^*-algebras A_n does every homomorphism $H \to \mathcal{U}(\prod_n A_n / \bigoplus_n A_n)$ still map w to the identity?

(3) For which groups do the criteria of Theorems S2.3, S2.7 and S6.1 compute the residual exactly?

S12. The maximal group C^*-algebra

For a countable group H, the maximal group C^*-algebra $C^*_{\max}(H)$ is the completion of the group ring $\mathbb{C}[H]$ in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of H; the supremum is finite because $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$. Write u_h for the canonical unitary of $h \in H$.

Proposition S12.1 (the universal property)**.** *The canonical map $h \mapsto u_h$ is injective, and for every unital C^*-algebra B and every homomorphism $\rho\colon H \to \mathcal{U}(B)$, there is a unique unital $*$-homomorphism $C^*_{\max}(H) \to B$ with $u_h \mapsto \rho(h)$ for every $h \in H$.*

Injectivity follows from the left regular representation, and uniqueness follows from density of $\mathbb{C}[H]$.

Proposition S12.2 (a strictly dominated projection gives a proper isometry)**.** *Let A be a unital C^*-algebra containing a projection p and a unitary u with $p < upu^*$. Then A contains an isometry that is not a unitary; consequently A is not stably finite and has no faithful tracial state.*

Proof. Write $q = upu^*$, so q is a projection with $pq = qp = p$, and put $r = pu^*$. Then

$$r^*r = up\,pu^* = upu^* = q, \qquad rr^* = pu^*up = p.$$

Put $s = r + (1-q)$. Since $p(1-q) = 0$ and $(1-q)p = 0$,

$$s^*s = r^*r + (1-q) = q + (1-q) = 1,$$

and since $pu^*(1-q) = pu^* - pu^*upu^* = pu^* - pu^* = 0$ and likewise $(1-q)up = 0$,

$$ss^* = rr^* + (1-q) = p + (1-q) = 1 - (q-p).$$

Thus $s^*s = 1$ but $ss^* = 1-(q-p) \neq 1$, so s is a nonunitary isometry. A unital C^*-algebra containing a nonunitary isometry is not finite, hence not stably finite. If τ were a faithful tracial state then $\tau(1-ss^*) = \tau(1)-\tau(s^*s) = 0$ while $1 - ss^* = q - p$ is a nonzero positive element, contradicting faithfulness. ☐

Proposition S12.3 (the maximal algebra under proper one-sided conjugation)**.** *Let G be a countable group containing a property-(T) subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ is not stably finite, has no faithful tracial state, and is in particular neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the spectral projection at 1 of the Hermitian Kazhdan average $\frac{1}{2|S|}\sum_{s\in S}(u_s + u_s^*)$, which exists by Lemma 3.4. Then $P \le u_t P u_t^*$. This inequality is strict. In the quasi-regular representation on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$. Thus the two fixed-space projections are distinct in this representation. Now apply Proposition S12.2. ☐

The base subgroup of E and the element t form such a pair, so $C^*_{\max}(E)$ is not stably finite. The reduced algebra behaves differently: its canonical trace is faithful and it is stably finite, despite being non-MF (Theorem C).

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AI and computational resource usage

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